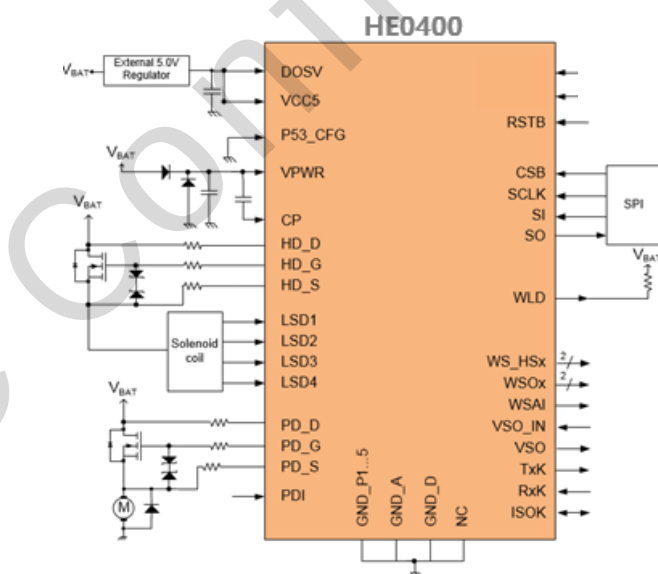


ABS motorcycle braking chip

The HE0400 is an antilock brake controller designed for use in harsh motorcycle environments. It has four high-current low-side drivers for use with solenoid valves, two configurable wheel speed sense inputs capable of handling active sensors, and high-side gate drivers for controlling two external N-channel MOSFETs for use with a pump motor and master relay for solenoid coils. Along with this main functionality, the HE0400 also has a warning lamp driver and a K-line transceiver. The HE0400 boosts an internal charge pump, allowing the high-side drivers to use inexpensive N-channel MOSFETs. The digital I/O pins can be configured for both 5.0 V and 3.3 V levels for easy connection to any microprocessor. The HE0400 uses standard SPI protocol for communication. The HE0400 is a solution for ABS braking application, and powered by SMARTMOS technology.

Features

- Four PWMed valves driver up to 5.0 A (5.0 kHz)
- High-side pre-driver for valves protection
- Two wheel speed sensor interfaces (active)
- Dual vehicle speed outputs
- Pump motor pre-driver up to 500 Hz PWM
- 16-bit SPI interface with watchdog
- K-line interface
- Warning lamp driver
- Die temperature warning
- Supervision



Part Number	Package	Marking	Packing
HE0400RNAA	QFN48	HE0400	2500/Tape&Reel

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

*The full part number will be confirmed after the official version of the product.

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1. Internal block diagram

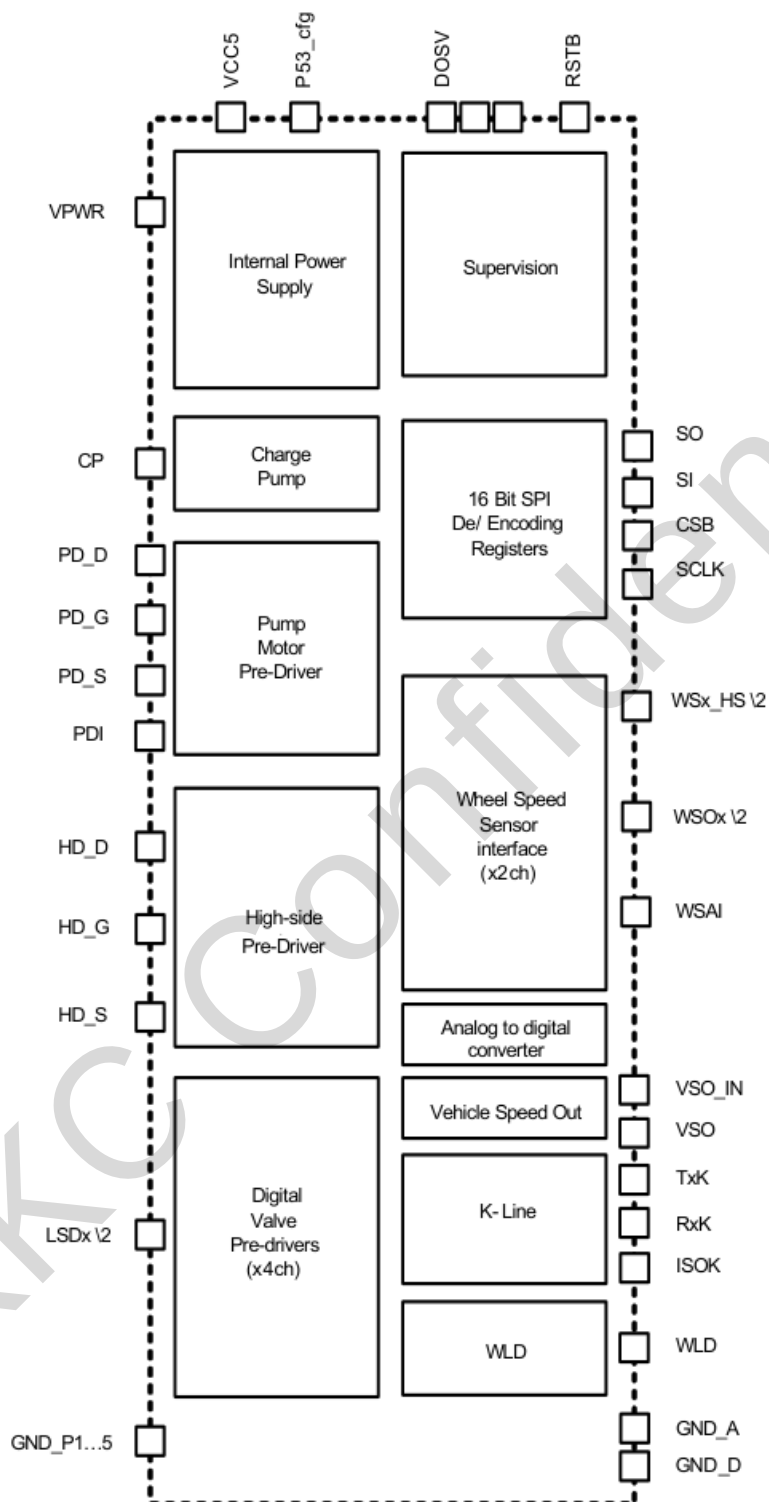


Figure 2. HE0400 simplified block diagram

2 Pin connections

2.1 Pinout diagram

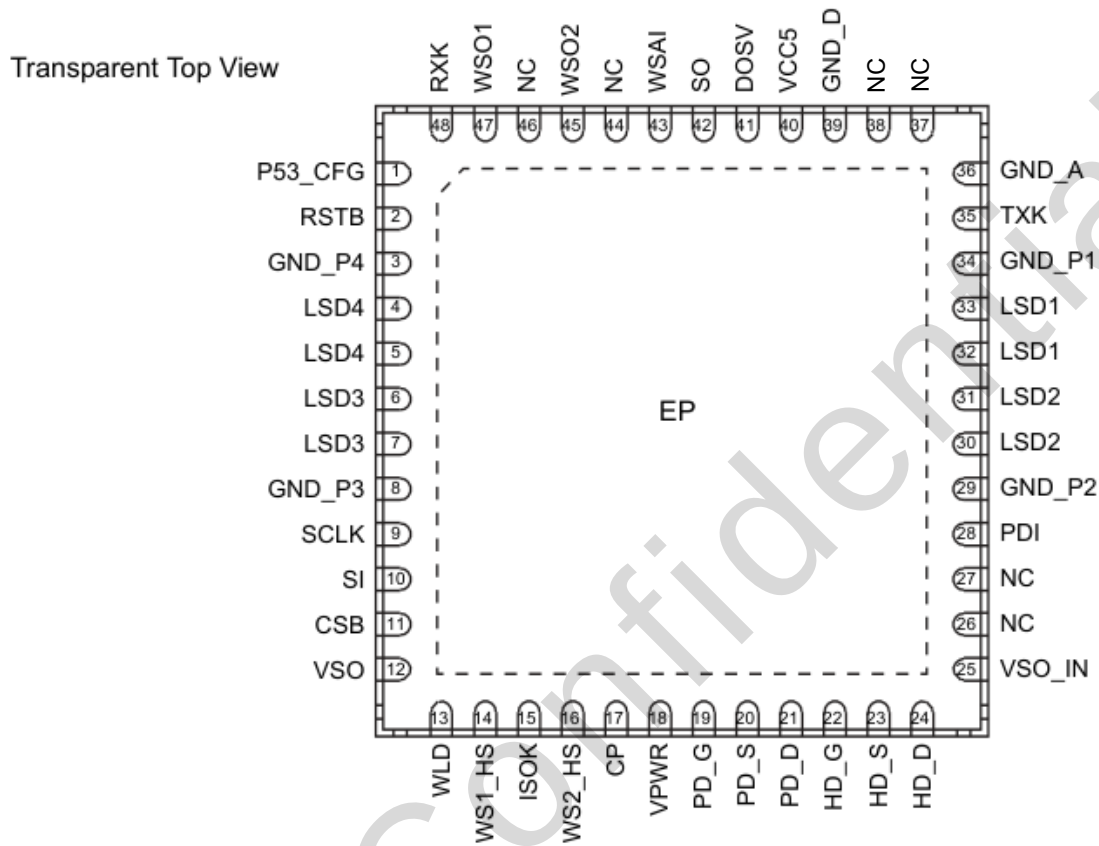


Figure 2-1 HE0400 48-Pin QFN-EP pinout diagram

2.2 Pin definitions

Table 2-1 HE0400 pin definitions

Pin number	Pin name	Pin function	Definition	DOSV = 5.0 V	DOSV = 3.3 V
1	P53_CFG	Supply	Input to select output voltage at DOSV (5.0 V/3.3 V)	no	no
2	RSTB	Reset	Reset pin	external pull-up	
3	GND_P4	Supply	Power ground 4 ⁽³⁾	no	no
4, 5	LSD4	Low-side Driver for Valves	Open drain output for low-side driver 4 ⁽¹⁾	no	no
6, 7	LSD3	Low-side Driver for Valves	Open drain output for low-side driver 3 ⁽¹⁾	no	no
8	GND_P3	Supply	Power ground 3 ⁽³⁾	no	no
9	SCLK	SPI	SPI interface clock input	no	no
10	SI	SPI	SPI interface digital input	no	no
11	CSB	SPI	SPI interface chip interface	no	no
12	VSO	Vehicle Speed Output	Vehicle speed output	no	no
13	WLD	Warning Lamp Driver	Open drain output for warning lamp driver	no	no
14	WS1_HS	Wheel Speed Sensor Interface	Sensor supply high-side. Supplies hall sensor: current limited, reverse protected supply	no	no
15	ISOK	K-LINE	K-line signal input/output for 12 V diagnostic	no	no
16	WS2_HS	Wheel Speed Sensor Interface	Sensor supply high-side. Supplies hall sensor: current limited, reverse protected supply	no	no
17	CP	Charge Pump	Charge pump output. For internal usage, connect a storage capacitor of 220 nF to VPWR.	no	no

Table 2-2 HE0400 pin definitions (continued)

Pin number	Pin name	Pin function	Definition	DOSV = 5.0 V	DOSV = 3.3 V
18	VPWR	Supply	Supply pin for HE0400. Connect to battery through reverse diode	no	no
19	PD_G	Motor Pump Driver	Gate output to control pump motor FET. Connect to gate of external pump motor FET	no	no
20	PD_S	Motor Pump Driver	Source feedback pump motor FET. Connect to source of external pump motor FET	no	no
21	PD_D	Motor Pump Driver	Drain feedback pump motor FET. Connect to drain of external pump motor FET	no	no
22	HD_G	High-side Driver for Valve's Fail Safe FET	Gate output to control high-side FET. Connect to gate of external high- side FET	no	no
23	HD_S	High-side Driver for Valve's Fail Safe FET	Source feedback high-side FET. Connect to source of external high-side FET	no	no
24	HD_D	High-side Driver for Valve's Fail Safe FET	Drain feedback high-side FET Connect to drain of external high-side FET	no	no
25	VSO_IN	Digital Input of VSO	Digital Input to control the vehicle speed output	no	no
28	PDI	Motor Pump Driver	Motor pump direct control (PDI_en should be set)	no	no
29	GND_P2	Supply	Power ground 2 (3)	no	no
30, 31	LSD2	Low-side Driver for Valves	Open drain output for low-side driver 2 (1)	no	no
32, 33	LSD1	Low-side Driver for Valves	Open drain output for low-side driver 1 (1)	no	no
34	GND_P1	Supply	Power ground 1 (3)	no	no
35	TXK	K-LINE	K-line logical input	no	no
36	GND_A	Supply	Analog ground	no	no
37	NC	-	-	no	no
38	NC	-	-	no	no
39	GND_D	Supply	Digital ground	no	no
40	VCC5	Supply	5.0 V supply pin for HE0400	5V	5V
41	DOSV	Supply	Digital output voltage supply, DOSV undervoltage reset	5V	3.3V
42	SO	SPI	SPI interface digital output	DOSV bias	
43	WSAI	Wheel Speed Sensor Interface	Digital output for additional information from sensor	DOSV bias	
45	WSO2	Wheel Speed Sensor Interface	Sensor output 2	DOSV bias	
47	WSO1	Wheel Speed Sensor Interface	Sensor output 1	DOSV bias	
48	RXK	K-LINE	K-line logical output	DOSV bias	
26, 27	NC	Not connected	Pin used for production tests and should be grounded	no	no
44, 46	NC	Not connected	Pin used for production tests and should not be grounded, remain opened	no	no
Exposed pad	GND_P5	Supply	Power ground 5 (3)	no	no

Notes:

1. Pins should be shorted together.
2. NA
3. All GND_Px pins should be shorted together at the PCB level.

3 General product characteristics

3.1 Maximum ratings

Table 3-1 Maximum ratings

Voltage parameters are absolute voltages referenced to GND_A, GND_D and the flag (tied together internally). Exceeding these ratings may cause a malfunction or permanent damage to the device.

Symbol	Description (rating)	Min.	Max.	Unit	Num
Supply					
VVPWR	Analog Power Supply Voltage	-0.3	40	V	P_3.1.1
VDOSV	Digital Output Supply Voltage	-0.3	7.0	V	P_3.1.2
VP53_CFG	Digital IO Voltage Selection for 5.0 V or 3.3 V	-0.3	7.0	V	P_3.1.3
VVCC5	Digital Power Supply Voltage	-0.3	7.0	V	P_3.1.4
VGND_A	Ground Analog	-0.3	0.3	V	P_3.1.5
VGND_D	Ground Digital	-0.3	0.3	V	P_3.1.6
VGND_P	Ground Exposed pad	-0.3	0.3	V	P_3.1.7
Charge pump					
VCP	Internal Charge Pump Voltage	-0.3 or VPWR - 0.3	VPWR +15	V	P_3.1.10
High-side driver for valve's fail-safe FET					
VHD_G	Gate Voltage of the High-side Pre-driver	-20	55	V	P_3.1.11
VHD_S	Source Voltage of the High-side Pre-driver	-20	40	V	P_3.1.12
VHD_D	Drain Voltage of the High-side Pre-driver	-0.3	40	V	P_3.1.13
Motor pump driver					
VPD_G	Gate Voltage of the Motor Pump Pre-driver	-0.3	55	V	P_3.1.14
VPD_S	Source Voltage of the Motor Pump Pre-driver	-0.3	40	V	P_3.1.15
VPD_D	Drain Voltage of the Motor Pump Pre-driver	-0.3	40	V	P_3.1.16
VPDI	Digital Motor Pump Input Voltage	-0.3	7.0	V	P_3.1.17
Reset					
VRSTB	Voltage Reset Pin	-0.3	7.0	V	P_3.1.18
SPI					
VSO	Serial Peripheral Interface Slave Output	-0.3	DOSV +0.3	V	P_3.1.19
VSI	Serial Peripheral Interface Slave Input	-0.3	7.0	V	P_3.1.20
VCSB	Serial Peripheral Interface Chip Select	-0.3	7.0	V	P_3.1.21
VSCLK	Serial Peripheral Interface Clock	-0.3	7.0	V	P_3.1.22

Table 3-2 Maximum ratings

Voltage parameters are absolute voltages referenced to GND_A, GND_D and the flag (tied together internally). Exceeding these ratings may cause a malfunction or permanent damage to the device.

Symbol	Description (rating)	Min.	Max.	Unit	Num
Wheel speed sensor interface					
VWSOx	Wheel Speed Sensor Digital Output Voltage	-0.3	DOSV +0.3	V	P_3.2.1
VWSAI	Wheel Speed Sensor Digital Output Voltage	-0.3	DOSV +0.3	V	P_3.2.2
VWSx_HS	Wheel Speed Sensor Analog Input Voltage	-0.3	40 or VPWR +0.3	V	P_3.2.3
VVSO_IN	Digital Vehicle Speed Input Voltage	-0.3	7.0	V	P_3.2.4
VVSO	Vehicle Speed Output Voltage	—	40	V	P_3.2.5
	Vehicle Speed Output Current	-100	—	mA	P_3.2.6
Low-side driver for valves (LSD1-4)					
VLSDX	Low-side Driver (for valves) Voltage	table 3-3	Active clamp		P_3.2.7
Warning lamp driver					
VWLD	Warning Lamp Driver Voltage	—	40	V	P_3.2.8
	Warning Lamp Driver Current	-100	—	mA	P_3.2.9
K-Line					
VISOK	Serial Link Interface Input/Output Voltage	—	40	V	P_3.2.10
	Serial Link Interface Input/Output Voltage Current	-100	—	mA	P_3.2.11
VRxK	Digital Serial Link Interface Input Voltage	-0.3	DOSV +0.3	V	P_3.2.12
VTXK	Digital Serial Link Interface Output Voltage	-0.3	7.0	V	P_3.2.13
Energy capability					
ELSD1—4	Energy Capability (EAR) at TA = 125 °C (6) LSD1—4, (tests setup 20 mH load)	—	TBD	mJ	P_3.2.14
Currents capability					
ILSDX(POS)	Drain Continuous Current; during on state LSDx	—	5.0	A	P_3.2.15
ILSDX(NEG)	Maximum Negative Current for 5.0 ms without being destroyed LSDx	-6.0	—	A	P_3.2.16
IDIG	Input Current P53_CFG, SI, CSB, SCLK, VSO_IN, TxK, RSTB	-20	20	mA	P_3.2.17
ESD protection					
ESDHBM	Human Body Model (4) All pins	-2.0	2.0	kV	P_3.2.18
	VPWR, WS_HSx, VSO, WLD, ISOK	-6.0	6.0		
ESDCDM	Charge Device Model (5) All pins Corner pins	-500 -750	500 750	V	P_3.2.19

Notes.

- Human Body Model (HBM): AEC-Q_100 Rev. G
- Charge Device Model (CDM): AEC-Q-100 Rev. G
- The energy level is confirmed with Silicon prototypes characterization.

3.2 Operating conditions

This section describes the operating conditions and the current consumptions. Conditions apply to all the following data, unless otherwise noted.

Table 3-3 Operating conditions

Voltage parameters are absolute voltages referenced to GND_A, GND_D, and the flag (tied together internally). Exceeding these ratings may cause a malfunction or permanent damage to the device.

Symbol	Characteristic	Min.	Typ.	Max.	Unit	Num
VPWR	Functional Operating Supply Voltage. Device is fully functional. All features are operating Extended operating supply voltage	6.0 5.3	— —	20 30	V	P_3.3.1
VCC5	Functional Operating Supply Voltage. Device is fully functional. (7) All features are operating	4.75	—	5.25	V	P_3.3.2
VDOSV	Functional Operating Supply Voltage. Device is fully functional. (8) All features are operating	3.13	—	5.25	V	P_3.3.3

Notes:

7. all parameters between 6.0 V to 20 V.
8. Between a maximum VCC5_UV value to a minimum VCC5 value, the device works in normal mode (A noise margin has to be take into account between the maximum VCC5_UV value to the minimum VCC5 value = 100 mV).

3.3 Supply currents

This section describes the operating conditions and the current consumptions. Conditions apply to all the following data, unless otherwise noted.

Table 3-4 Supply currents

Characteristics noted under conditions $6.0\text{ V} \leq V_{PWR} \leq 20\text{ V}$, $4.75\text{ V} \leq V_{CC5} \leq 5.25\text{ V}$, $3.13\text{ V} \leq V_{DOSV} \leq 5.25\text{ V}$, $-40\text{ }^{\circ}\text{C} \leq T_J \leq 150\text{ }^{\circ}\text{C}$, GND = 0 V, unless otherwise noted. Typical values noted reflect the approximate parameter means at $T_A = 25\text{ }^{\circ}\text{C}$ under nominal conditions, unless otherwise noted.

Symbol	Characteristic	Min.	Typ.	Max.	Unit	Num
VPWR current consumptions						
IQVPWR	Sleep Mode Measured at VPWR = 14 V	—	3.0	30	mA	P_3.4.1
IVPWR	Operating Mode (WSx_HS No load)	—	10	20	mA	P_3.4.2
VCC5 current consumptions						
IVCC5	Operating Mode	—	10	20	mA	P_3.4.3
DOSV current consumptions						
IDOSV	Operating Mode (SPI frequency 10 MHz)	—	—	10	mA	P_3.4.4

3.4 Thermal data

Table 3-5 Thermal data

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
TJ	Operational Junction Temperature (9)	-40	—	150	°C	P_3.5.1
TSTG	Storage Temperature	-65	—	150	°C	P_3.5.2
RθJC-BOTTOM	RθJC, Thermal Resistance, Junction to Case Bottom (Package exposed pad) - Steady state (10)	—	—	0.7	°C/W	P_3.5.3
TPPRT	Peak Package Reflow Temperature During Reflow (11)	—	—	260	°C	P_3.5.4

Notes

9. The total device lifetime at a junction temperature of 150 °C should be guaranteed for 1000 hours.
10. Thermal characterization parameter indicating the temperature difference between package bottom center and the junction temperature per JEDEC JESD51-12.
11. Pin soldering temperature limit is for 10 seconds maximum duration. Not designed for immersion soldering. Exceeding these limits may cause malfunction or permanent damage to the device.

3.5 Logical inputs/outputs

Table 3-6 Logical inputs/outputs

VPWR = 6.0 to 20 V, VCC5 = 4.75 to 5.25 V, DOSV = 3.13 to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Characteristic	Min.	Typ.	Max.	Unit	Num
Logical inputs						
VIH_X	Input High-voltage P53_CFG, RSTB, SI, CSB, SCLK, VSO_IN, TxK	—	—	2.0	V	P_3.6.1
VIL_X	Input Low-voltage P53_CFG, RSTB, SI, CSB, SCLK, VSO_IN, TxK	0.8	—	—	V	P_3.6.2
VHYS_X	Input Voltage Threshold Hysteresis P53_CFG, RSTB, SI, CSB, SCLK, VSO_IN, TxK	100	—	—	mV	P_3.6.3
IPD_X	Pin Pull-down Current, with VINPUT = 1.0 V VSO_IN, SI, SCLK, PDI (if PDI_en = 1)	5.0	—	20	uA	P_3.6.4
IPU_X	Pin Pull-up Current, with VINPUT = 2.0 V TxK, CSB	5.0	—	20	uA	P_3.6.5
RPD_X	Pull-down Resister; Driver Off State RSTB, P53_CFG	100	—	400	kΩ	P_3.6.6
CIN	Input Capacitance All digital inputs	—	—	12	pF	P_3.6.7
Logical outputs						
VOH_X	Output High-voltage, with 1.0 mA SO, WSOx, WSAI, RxK	0.8xDOSV	—	—	V	P_3.6.8
VOL_X	Output Low-voltage, with 1.0 mA SO, WSOx, WSAI, RxK	—	—	0.4	V	P_3.6.9
ISO_LK	SO Tri-state Leakage Current SO	-5.0	—	5.0	uA	P_3.6.10
VOL_RSTB	RSTB Low-voltage, with 1.0 mA RSTB	—	—	0.4	V	P_3.6.11

3.6 Block diagram

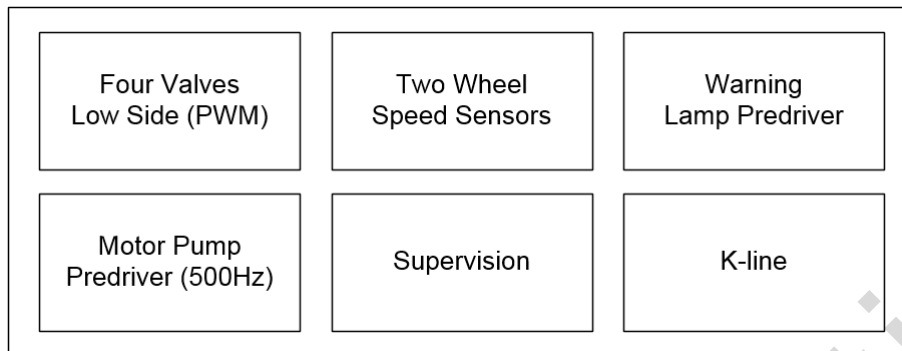


Figure 3-1 HE0400 functional block diagram

3.7 Functional description

The HE0400 device is an antilock brake controller designed for use in harsh motorcycle environments, requiring few external components. It has four high-current low-side drivers for use with solenoid valves, two configurable wheel speed sense inputs capable of handling both Intelligent and active sensors, and high-side gate drivers for controlling two external N-channel MOSFETs for use with a pump motor and master relay for a solenoid coil. In conjunction with this primary functionality, the HE0400 has warning lamp pre-drivers and a K-line transceiver.

The HE0400 boosts an internal charge-pump, allowing the high- side drivers to use inexpensive N-channel MOSFETs. The digital I/O pins can be configured for both 5.0 V and 3.3 V levels for easy connection to any microprocessor. The HE0400 uses standard SPI protocol for communication.

3.8 Features

This section presents the detailed features of HE0400.

Table 3-7 Device features set

Function	Description
High-side Driver for Fail-safe FET	• High-side fail-safe FET driver • Overcurrent shutdown • Load leakage detection
Pump Driver	• Pump motor driver up to 500 Hz PWM frequency • Direct control through the PDI pin • Overcurrent shutdown between external FET Drain and Source • Load dump activation
Low-side Solenoid Driver (x4)	• Solenoid driver (160 mΩ max RDS(on) at 150 °C) are PWM low-side driver • Configurable PWM frequency from up to 5.0 kHz • PWM duty cycle resolution 0.39% • Open load detection • VDS state monitoring • Overcurrent shutdown • Overtemperature shutdown • Max switch-off energy 56 mJ

Wheel Speed Sensor I/F (x2)	• Wheel Speed sensor power supply (high-side) • Open load detection • Overcurrent shutdown • Overtemperature shutdown • Overvoltage regulation • The sensor types supported are standard active two-level wheel speed sensors (7/14 mA), smart active two-level wheel speed sensors (7/14 mA), and VDA type three-level wheel speed sensors (7/14/ 28 mA)
-----------------------------	--

Table 3-8 Device features set (continued)

Function	Description
VSO	• Vehicle speed out1 • Open load detection • VDS state monitoring • Overcurrent shutdown • Overtemperature shutdown
K-line	• ISO9141K-line interface or vehicle speed out2 • Overcurrent shutdown • Overtemperature shutdown • Open load detection
WLD	• Warning Lamp driver1 (20 mA max.) • Open load detection • VDS state monitoring • Overcurrent shutdown • Overtemperature shutdown
Supervision	• VINT_x undervoltage (internal regulator) • VCC5 & DOSV undervoltage (supply voltage from external) • Watchdog fault • ALU check counter overflow • External reset fault • VPWR Undervoltage and overvoltage detections • Mismatch MAIN-AUX OSC CLK • Temperature warning • SPI failure • Charge pump error • GND supervision

4 Functional block description

4.1 Error handling

Table 4-1 Error handling

Type of error	Detection condition	Action	Clear SPI flag	Restart condition
High-side driver				
Overcurrent between external FET Drain and Source	ON	HD_G Off + SPI fault flag (HD_oc)	Write 1 to HD_clr_fit 1	Write 1 to HD_clr_fit and then turn on by SPI command (hd on)
Load leakage	hd_on rise-edge (SPI bit)	Ignore hd_on rise-edge command + SPI fault flag (HD_lkg)	Write 1 to HD_clr_fit	Write 1 to HD_clr_fit and then turn on by SPI command (hd on)
Pump driver				
Overcurrent between external FET Drain and Source	ON	PD_G Off+ SPI fault flag (PD_oc)	Write 1 to PD_clr_fit	Write 1 to PD_clr_fit and then turn-on by SPI command (pd on)
LSDx				
Open load	OFF	SPI flag only (LSDx_op)	Read diagnosis	No
VDS state monitoring	ON/OFF	Send VDS state by SPI (vds_LSDx)	Update with min filter time (T1) rise and fall edge	No
Overcurrent	ON	OFF fault FET only + SPI fault flag (LSDx_oc)	Write 1 to LSD_clr_fit	Write 1 to LSD_clr_fit and then turn on by SPI command (LSDx duty cycle)
Overtemperature	ON	OFF fault FET only + SPI fault flag (LSDx_ot)	Write 1 to LSD_clr_fit	Write 1 to LSD_clr_fit and then turn on by SPI command (LSDx duty cycle)
Wheel speed sensor WSx_HS				
Open load/short to battery	ON	SPI flag only (WSx_op)	Read diagnosis	No
Overcurrent	ON	OFF fault FET only + SPI fault flag (WSx_oc)	Write 1 to ws_clr_fit	Write 1 to WS_clr_fit and then turn on by SPI command (WSx_dis)
Overtemperature	ON	OFF fault FET only + SPI fault flag (WSx_ot)	Write 1 to ws_clr_fit	Write 1 to WS_clr_fit and then turn on by SPI command (WSx_dis)
Overvoltage regulation	ON	No	No	No
WSx leakage fault	ON	SPI flag only (WSx_lkg)	Read diagnosis	No
VSO				
Open load	OFF	SPI flag only (VSO_op)	Read diagnosis	No
VDS state monitoring	ON/OFF	Send VDS state by SPI (vds_VSO)	Update with min filter time (T1) rise and fall edge	No
Overcurrent	ON	OFF fault FET only + SPI fault flag (VSO_oc)	Write 1 to vso_clr_fit	Write 1 to VSO_clr_fit and then turn on automatically; follow previous SPI command (WS01 or VSO IN)
Overtemperature	ON	OFF fault FET only + SPI fault flag (VSO_ot)	Write 1 to vso_clr_fit	Write 1 to VSO_clr_fit and then turn on automatically; follow previous SPI command (WS01 or VSO IN)

K-Line

Overcurrent	ON	OFF fault FET only + SPI fault flag (ISOK_oc)	Write 1 to ISOK_clr_fit	Write 1 to ISOK_clr_fit and then turn on automatically; followTxK
Overtemperature	ON	OFF fault FET only + SPI fault flag (ISOK_ot)	Write 1 to ISOK_clr_fit	Write 1 to ISOK_clr_fit and then turn on automatically; followTxK

Table 4-2 Error handling (continued)

Type of error	Detection condition	Action	Clear SPI flag	Restart condition
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WLD

Open load	OFF	SPI flag only (WLD_op)	Read diagnosis	No
VDS state monitoring	ON/OFF	Send VDS state by SPI (vds_WLD)	Update with min. filter time (T1) rise and fall edge	No
Overcurrent	ON	OFF fault FET only + SPI fault flag (WLD_oc)	Write 1 to WLD_clr_fit	Write 1 to WLD_clr_fit and then turn on by SPI command (WLD_on)
Overtemperature	ON	OFF fault FET only + SPI fault flag (WLD_ot)	Write 1 to WLD_clr_fit	Write 1 to WLD_clr_fit and then turn on by SPI command (WLD_on)

Supervision

VINT_x undervoltage	All except sleep mode	See Table 4-28	See Table 4-28	See Table 4-28
VCC5 & DOSV undervoltage	All except sleep mode	See Table 4-28	See Table 4-28	See Table 4-28
Watchdog fault	RSTB is high state	See Table 4-28	See Table 4-28	See Table 4-28
ALU check counter overflow	RSTB is high state	See Table 4-28	See Table 4-28	See Table 4-28
External reset fault	No internal RSTB pull-down	See Table 4-28	See Table 4-28	See Table 4-28
VPWR Undervoltage	RSTB is high state	All LSDx Off (Clear of All LSDx duty cycle registers or current set point) + SPI fault flag (VPWR_UV) + Ignore LD- (See Table 4-28)	Normal condition ⁽¹²⁾ Read diagnosis ⁽¹³⁾ (VPWR_UV) (See Table 4-28)	Normal condition ⁽¹³⁾ Turn on by SPI ⁽¹³⁾ command (LSDx duty cycle or current set point). (See Table 4-28)
VPWR Overvoltage	RSTB is high state	All LSDx Off (Clear of All LSDx duty cycle registers or current set point) + SPI fault flag (VPWR_OV) + HSD shutdown (delay time between HSD shutdown & LSD shutdown + PD_G turn on during tLDact; off with normal condition with a certain filter time (See Table 4-28))	Normal condition ⁽¹²⁾ Read diagnosis ⁽¹³⁾ (VPWR_OV) (See Table 4-28)	Normal condition ⁽¹²⁾ Turn on by SPI command ⁽¹³⁾ (LSDx duty cycle or current set point). (See Table 4-28)
Load dump	RSTB is high state	PD_G turn on during tLDact; off with normal condition with a certain filter time	Read diagnosis	No

Mismatch MAIN- AUX OSC CLK	RSTB is high state	See Table 4-28	See Table 4-28	See Table 4-28
Temperature warning	RSTB is high state	SPI flag only	Normal condition ⁽¹²⁾ Read diagnosis ⁽¹³⁾	No
SPI failure	RSTB is high state	SPI flag only (Fmsg)	Read diagnosis	No
GND_D supervision	RSTB is hi gh state	SPI flag only (FGND)	No	No
GND_A supervision; indirect detection by VCC5 or DOSV	RSTB is high state	SPI flag only (VCC5_UV or DOSV_UV)	No	No

Notes

12. If xxx_clr_flg is written "1" by the SPI, the affected SPI flags are clear to "0", so the SW engineer has to read SPI flag first and then write xxx_clr_flg carefully default to "0".
13. Internal voltages can be monitored in real time due to SPI reading, and can use the fail-safe function. No power FET turn on during un-power mode (VPWR = 0 and VINT_x = 0); WLD, VSO, ISOK, LSDx.

4.2 High-side driver

4.2.1 Function description

The high-side driver is intended to control the fail-safe switch for the overall solenoid path, and HD_G is controlled by the SPI command. For reverse protection by the FET application, the HD_G has no leakage current in Sleep mode (BAT = 14 V, VCC5 = DOSV = 0 V)

For reverse protection by the diode application, the HD_D has no leakage current in Sleep mode (BAT = 14 V, VCC5 = DOSV = 0V)

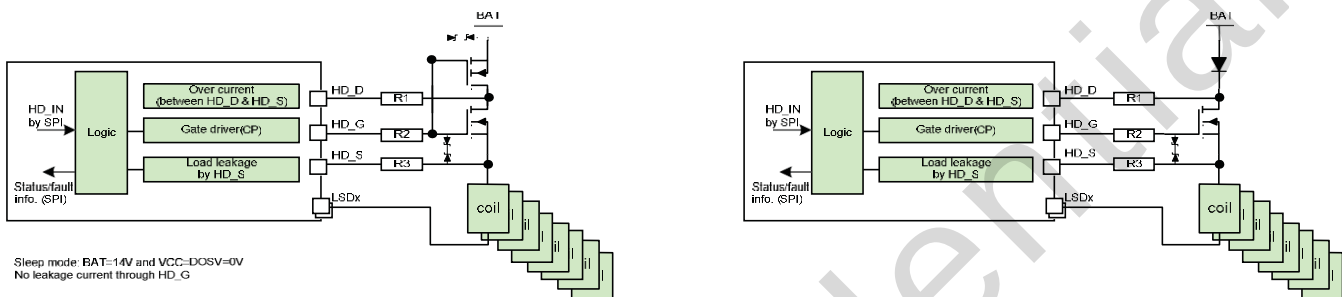


Figure 4-1 High-side driver

4.2.2 High-side driver and fault protection

4.2.2.1 Overcurrent

High-side driver protects the external n-channel power FET on HD_G in overcurrent conditions. The drain-source voltage of the FET on HD_G is checked if the high-side driver is switched on. If the measured drain-source voltage exceeds the overcurrent voltage threshold, the output of the overcurrent comparator is enabled. If the output of the comparator is active longer than the defined filter time, the output HD_G is switched off. Overcurrent detection logic has a masking time from hd_on turn-on against malfunctions on transient time. After switching off the power FET on HD_G by an overcurrent condition, the power FET can be turned back to a “normal state” by a SPI write 1 to the “HD_clr_flg” register, and then turned on by a SPI command.

4.2.2.2 Load leakage detection

Each time HD_G is turned on, the ILCdet current is sourced out of the HD_S pin for the time tHD_LC, to check the external leakage current on the node in the application. The high-side switch on HD_G is turned on if the measured voltage is over the detection threshold. If this test fails, HD_G does not turn-on and the fault flag is set to high. The power FET can be turned back to a “normal state” only by a SPI write 1 to the “HD_clr_flg” register, and then turned on by a SPI command. When the power FET is switched off, the gate capacitance of the FET is discharged by a constant current, which is controlled fast and slow by a SPI command (HPD_sr).

4.2.2.3 External components of high-side driver

For protection, external resistors RHD_D, RHD_G, and RHD_S are required (for example: RHD_D = 100 Ω, RHD_G = 100 Ω, RHD_S = 100 Ω). The zener clamping is necessary to protect the gate and source. The zener chains are used for avalanche clamping and protection against transients.

Table 4-3 High-side driver electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, TJ = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
HD_G						
VHD_ON	HD_G switch-on voltage - with pd_on: PWMat 500 Hz, 50% duty cycle thru the SPI, 16 nF between PD_G & Gnd. hd_on = 1 without PWM 5.5 V ≤ VPWR < 6.0 V 6.0 V ≤ VPWR < 7.0V 7.0 V ≤ VPWR < 10 V 10 V ≤ VPWR < 20 V	VPWR+ 4 VPWR+5 VPWR+7 VPWR+10	— — — —	VPWR+ 15 VPWR+15 VPWR+15 VPWR+15	V	P_4.3.1 P_4.3.2 P_4.3.3 P_4.3.4
VHD_OFF	HD_G switch-off voltage - pull-up current < 20 μA	—	—	0.8	V	P_4.3.5
tHD_ON	Turn-on time - After tHD_LC Test mode:16 nF load, RHDG = 100 Ω VPWR = 14 V, From digital input to VHDG = VPWR +5.0 V, 32 nF with two external FET	—	—	1.4	ms	P_4.3.6
IHD_OFF_SLOW	Turn-off current slow - HDG > 2.0 V. HPD_sr = 0	55	100	230	μA	P_4.3.7
IHD_OFF_FAST	Turn-off current fast - HDG > 2.0 V, HPD sr = 1	1.0	2.0	4.5	mA	P_4.3.8
tLSDX_HD_G	Delay time of HD_G shutdown after LSD shutdown due to supply fault or reset condition. - (See Table5.12-1) (reset table)	1.0	1.5	2.0	ms	P_4.3.9
RIHD_OFF	Turn-off resistance in sleep mode - VCC5 = DOSV = 0 V; HD_G = 1.0 V, HD_D = PD_D = VPWR = 14 V	—	0.2	12	kΩ	P_4.3.10
RHD_REV	Negative voltage resistance of HD_G - guaranteed by design, at HD_G to GND = -18 V	TBD	—	—	MΩ	P_4.3.11

HD_S

ILEAK_HD_SRC	Leakage current - 0 ≤ VHD_S ≤ 20 V, 6.0 ≤ VPWR ≤ 20 V	—	—	50	uA	P_4.3.12
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HD_D

LLEAK_HD_DRN	Leakage current - HD_D = PD_D = VPWR = 14 V VCC5 = DOSV = 0 V VCC5 = DOSV = 5 V, HDS=0V	— 105	— —	10 145	uA	P_4.3.13
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Overcurrent detection

VHD_OC	Overcurrent detection threshold - VHD_D-VHD_S, RDRN, RSRC = 100 Ω	-15%	1.0	+15%	V	P_4.3.14
tHD_OC	Overcurrent detection filter time	—	T2	—	us	P_4.3.15
MtHD_OC	Overcurrent detection masking time after turn ON - After tHD_LC	1.7	2.3	2.7	ms	P_4.3.16

Load leakage current detection

IHD_LC	HD_S source current	1.0	—	2.0	mA	P_4.3.17
VHD_LC	Leakage voltage threshold	3.0	3.5	4.0	V	P_4.3.18
tHD_LC	Load leakage current detection duration	—	2xT2	—	us	P_4.3.19
tHD_LC_fit	Load leakage current detection filter time	—	T2	—	us	P_4.3.20

4.3 Pump motor pre-driver

4.3.1 Function description

This module is designed for pump motor pre-drivers, a maximum of 500 Hz PWM is possible. The pump motor pre-driver can be driven by a SPI command (PD_on) and also by the external pin PDI by selecting the bit PDI_en = 1. When the PDI pin is selected and opened (no connection), the PD_G pin go to the default state (low state).

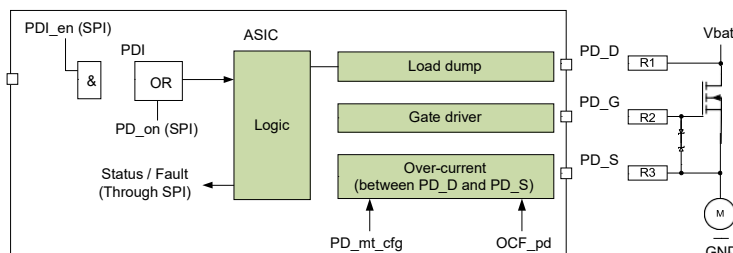


Figure 4-2 Pump motor pre-driver

4.3.2 Pump motor pre-driver control by SPI command

The motor pump pre-driver is controlled through the serial peripheral interface communication (SPI). To generate a duty cycle, the microcontroller has to send the bit pd-on as shown in [Figure 4-3](#).

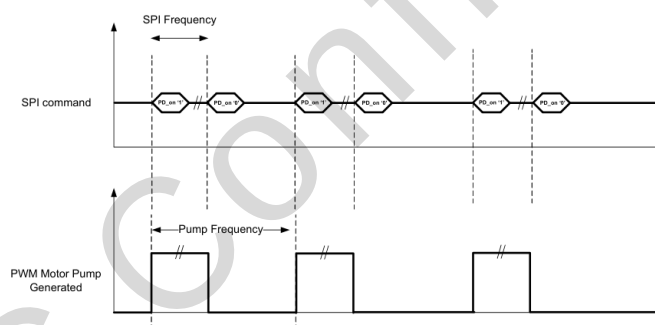


Figure 4-3 Pump motor pre-driver control

The duty cycle range allowed according to a certain frequency is representing in [Table 4-4\(14\)](#):

Table 4-4 Typical duty cycle (%) vs motor frequency (Hz)

Motor frequency (Hz)	Duty cycle (%)	
	Minimum value	Maximum value
100	7.24	92.76
200	14.48	85.52
300	21.72	78.28
400	28.96	71.04
500	36.2	63.8

Notes

14. This table was calculated with VBAT = 14 V, lpd_off_fast bit value = 1.0 mA, capacity MOSFET = 16 nF, SPI frequency = 10 MHz.

4.3.3 Fault detection

4.3.3.1 Overcurrent

The pump pre-driver protects the external n-channel power FET on PD_G in overcurrent conditions. The drain-source voltage of the FET on PD_G is checked if the high-side pre-driver is switched on. If the measured drain-source voltage exceeds the overcurrent voltage threshold, the output of the overcurrent comparator is enabled. If the output of the comparator is active longer than the defined filter time, the output PD_G is switched off. Overcurrent detection logic has a masking time from pd_on turn-on against malfunctions in transient time. The masking time and filter time of the pump pre-driver is controllable by the SPI bit (See [SPI and data register](#)). After switching off the power FET on PD_G by an overcurrent condition, the power FET can be turned back to a “normal state” by a SPI write 1 to the “PD_clr_fit” register, and then turned on by a SPI command.

When the power FET is switched off, the gate capacitance of the FET is discharged by a constant current, which is controlled fast and slow by a SPI command (HPD_sr).

4.3.3.2 External components of pump pre-driver

Protection of the resistors RPD_D, RPD_G, and RPD_S is required (for example: RPD_D = 2.0 kΩ, RPD_G = 100 Ω, RPD_S = 2.0 kΩ). Zener clamping is necessary to protect the gate and source. The zener chains are used for avalanche clamping and protection against transients.

Table 4-5 Pump motor pre-driver electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, TJ = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
PD_G						
VPD_ON	PD_G switch-on voltage - with pd_on: PWMat 500 Hz, 50% duty cycle thru the SPI, 16 nF between PD_G & Gnd. pd_on = 1 without PWM $5.5\text{ V} \leq \text{VPWR} < 6.0\text{ V}$ $6.0\text{ V} \leq \text{VPWR} < 7.0\text{ V}$ $7.0\text{ V} \leq \text{VPWR} < 10\text{ V}$ $10\text{ V} \leq \text{VPWR} < 20\text{ V}$	VPWR+ 4 VPWR+5 VPWR+7 VPWR+10	— — — —	VPWR+15 VPWR+15 VPWR+15 VPWR+15	V	P_4.5.1 P_4.5.2 P_4.5.3 P_4.5.4
VPD_OFF	PD_G switch-off voltage - pull-up current < 20 A	—	—	0.8	V	P_4.5.5
tPD_ON	Turn-on time - Test mode: 16 nF load, RPD_G = 100 Ω VPWR = 14 V, From digital input to VPD_G = VPWR +5.0 V	—	0.3	0.5	ms	P_4.5.6
IPD_OFF_SLOW	Turn-off current slow - VPD_G > 2.0 V. HPD_sr = 0	55	100	230	uA	P_4.5.7
IPD_OFF_FAST	Turn-off current fast - VPD_G > 2.0 V, HPD_sr = 1	1.0	2.0	4.5	mA	P_4.5.8
PD_S						
I _{LEAK_PD_S}	Leakage current - $0 \leq \text{VPD}_S \leq 20\text{ V}$, $6.0 \leq \text{VPWR} \leq 20\text{ V}$, VCC5=DOSV=0V	—	—	10	uA	P_4.5.9
PD_D						
I _{LEAK_PD_D}	Leakage current HD_D = PD_D = VPWR = 14 V, PD_S=0V VCC5 = DOSV = 0 V VCC5 = DOSV = 5 V	— 60	— —	10 85	uA	P_4.5.10 P_4.5.11

Overcurrent detection

VPD_OC	Overcurrent detection threshold - VPD_D-VPD_S, RPD_D, RPD_S = 2.0 k $\wedge$	-15%	1.0	+15%	V	P_4.5.12
tPD_OC1	Overcurrent detection filter time - OCF_pd = 0	—	T2	—	us	P_4.5.13
tPD_OC2	Overcurrent detection filter time - OCF_pd = 1	—	4*T1	—	us	P_4.5.14
MtPD_OC1	Overcurrent detection masking time after turn ON - 16 nF cap. between gate to source, PD_mt_cfg = 0, SPI command, default value = 0	0.8	1.1	1.4	ms	P_4.5.15
MtPD_OC2	Overcurrent detection masking time after turn ON - PD_mt_cfg = 1, SPI command	No masking time			ms	P_4.5.16

4.4 Low-side driver

4.4.1 Function description

The HE0400 is designed to drive inductive loads in low-side configurations. All four channels are monitored by logic, and faults are individually reported by the SPI. All external wiring to the loads and supply pins of the device are controlled. The device is self-protected against short-circuit and overtemperature at the outputs.

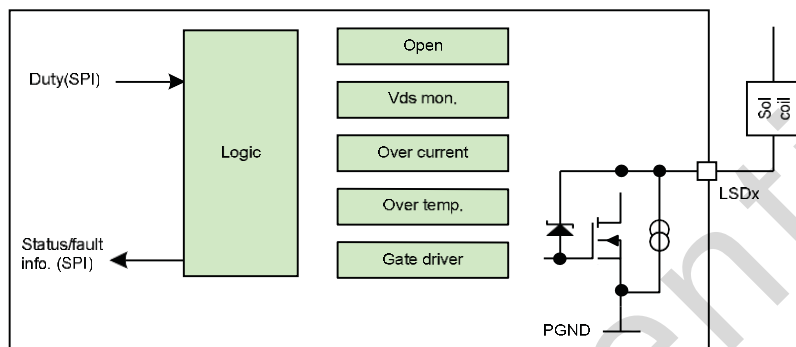


Figure 4-4 PWM low-side driver

Channels 1 to 4 are PWM low-side drivers. These outputs are composed of an output transistor, a pre-driver circuit, and diagnostic circuitry. The drivers can be controlled by the SPI control register.

The duty cycle of PWM low-side drivers is programmed via an 8-bit SPI message. Duty cycle between 0% and 100% can be selected and the LSB of the 8 bits is weighted with a 0.39% duty. Each channel has an 8-bit SPI register of the PWM duty cycle.

The PWM low-side driver can use each channel as a digital low-side switch.

PWMx duty cycle = 1111 1111 Digital low-side switch ON (conducting)

PWMx duty cycle = 0000 0000 → Digital low-side switch OFF

The HE0400 provides interleaved phase shift switching, to minimize switching noise of the solenoid coil.

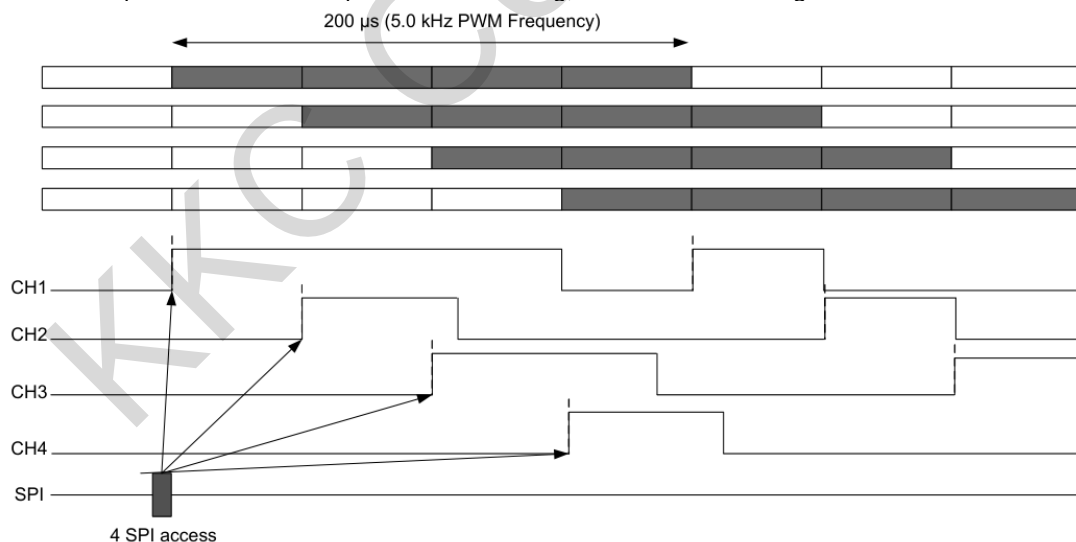


Figure 4-5 PWM valve control interleave

Table 4-6 Low-side driver electrical characteristics

VPWR = 6.0 V to 20 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
$R_{ON_LSD1\ to\ 4}$	On resistance channel 1 to 4, PWM - T_J = 150 °C, 9.0 V $\leq$ VPWR $\leq$ 20 V, I_{LOAD} = 2.0 A	—	—	0.160	Ω	P_4.6.1
I_{LEAK_LSD}	Drain leakage current - LSD = 14 V, sink current, disable with test mode	—	—	10	μ A	P_4.6.2
V_{CL_LSD}	Active clamp voltage	35	—	45	V	P_4.6.3
V_{BVDSS_LSD}	BVDSS voltage - I_{LEAK} < 1.0 mA	$V_{CL_LSD} + 1$	—	—	V	P_4.6.4

Timings

t_{D_ON}	Turn-on time Digital to 10% or 90%, 10% to 90%, I_{LOAD} = 1.0 A, VPWR = 14 V, no capacitor ⁽¹⁵⁾	0.0	—	2.5	μ s	P_4.6.5
t_{D_OFF}	Turn-off time Digital to 10% or 90%, 10% to 90%, I_{LOAD} = 1.0 A, VPWR = 14 V, no capacitor ⁽¹⁵⁾	0.0	—	4.0	μ s	P_4.6.6
LF_PWM	Output PWM frequency LF_PWM_14 = 111 LF_PWM_14 = 110 LF_PWM_14 = 101 LF_PWM_14 = 100 LF_PWM_14 = 000 (default) LF_PWM_14 = 011 LF_PWM_14 = 001 LF_PWM_14 = 010	-20%	3.0 3.2 3.4 3.6 3.9 4.2 4.5 5.0	+20%	kHz	P_4.6.7
00000000 00000001 ... 11111110 11111111	PWM Duty cycle Programming (8bits) - Can be used for digital low-side driver	— — — — —	OFF 0.39 ... 99.61 ON	— — — — —	%	P_4.6.8

Notes

15. Internal digital signal delivered by interleave synchronization block. See [PWM valve control interleave](#).

4.4.2 Fault detection**4.4.2.1 Open load**

An open condition is detected when the LSDx output is below the threshold for the defined filter time. The fault bit is set (SPI error flag only). This function is only operating during the off state.

4.4.2.2 V_{DS} state monitoring

The V_{DS} state monitoring gives a real time state of LSD drain voltage vs OP_LSD voltage. This signal is filtered and sent through the SPI. If the LSDx voltage is higher than OP_LSD with a filter time (T1), vds_lsd is set to "1".

4.4.2.3 Overcurrent

When the current is above the overcurrent threshold for the defined filter time, the driver is switched off, an SPI fault bit is set. The driver can be turned back to a "normal state" by a SPI write "1" to "LSDx_clr_fit", and then send a target current command.

4.4.2.4 Overtemperature

When the temperature is above the overtemperature threshold for the defined filter time, the driver is switched off, a SPI fault bit is set, and the turn-on SPI command is cleared. The driver can be turned back to a “normal state” when the temperature is returns to a normal state, a SPI write “1” to “LSDx_clr_fit”, and then a send target current command.

Table 4-7 Fault detection (LSDx) electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, $T_J = -40\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Overcurrent shutdown						
OC _{LSD}	Overcurrent detection threshold current	6.0	8.5	11	A	P_4.7.1
	$T_J = -40\text{ }^{\circ}\text{C}$	5.0	8.5	11		P_4.7.2
	$T_J = 25\text{ }^{\circ}\text{C}$	5.0	8.5	11		P_4.7.3
	$T_J = 150\text{ }^{\circ}\text{C}$					
t _{oCLSD}	Overcurrent detection filter time	—	T1	—	μs	P_4.7.4
Open load detection						
OP _{LSD}	Open Load detection threshold (also used for VDS monitoring)	1.6	2.0	2.4	V	P_4.7.5
I _{SINK_LSD}	Sink current - LSDx = 2 V	30	—	80	μA	P_4.7.6
t _{oP_LSD}	Open Load detection filter time	—	T2	—	μs	P_4.7.7
VDS monitoring						
t _{vDS_LSDX}	VDS state filter time - used Open Load detection comparator rise and fall edge filter time	—	T1	—	μs	P_4.7.8
Overtemperature shutdown						
OT _{LSD}	Overtemperature detection threshold	180	195	210	°C	P_4.7.9
t _{oT_LSD}	Overtemperature detection filter time	—	T1	—	μs	P_4.7.10

4.5 Load characteristic for low-side driver

This section helps the customer to define the maximum valve switching current which can be drained by the device in according to the maximum energy capability. Figure 4-6 presents the typical inductive switching and also the formulas to quantify the energy through the low-side driver (LSDx).

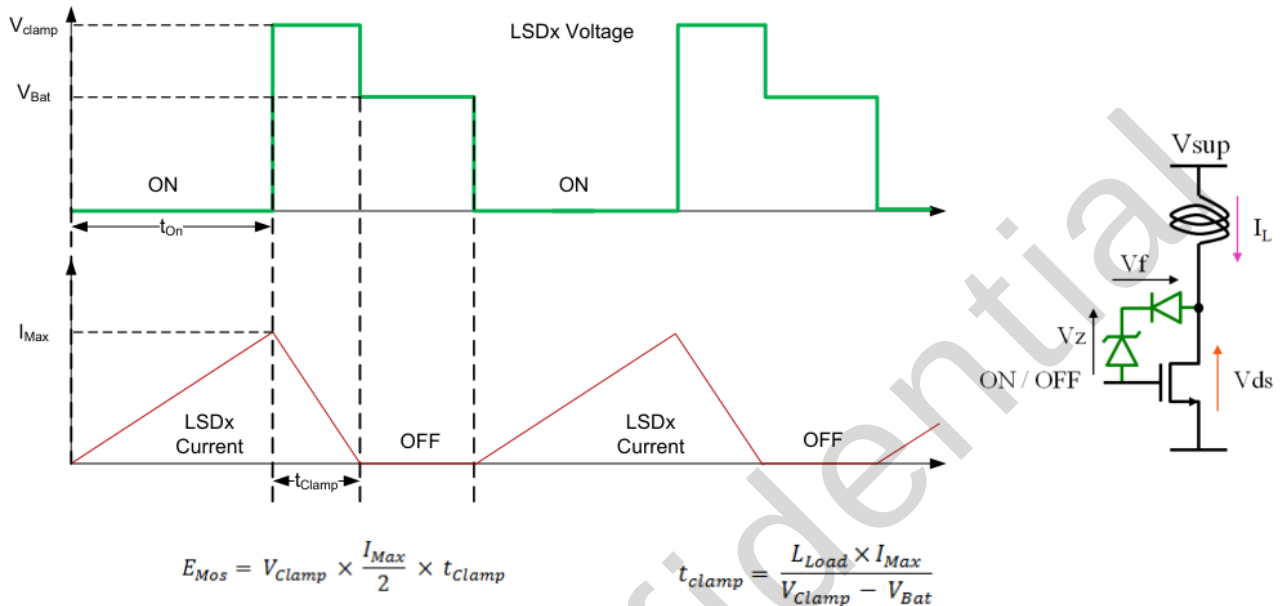


Figure 4-6 LSDx inductive switching

For reliability purpose, it is mandatory to satisfy: $E_{MOS} < E_{LSD1-4}$. The Figure 4-7 illustrates the Safe Operating Area:

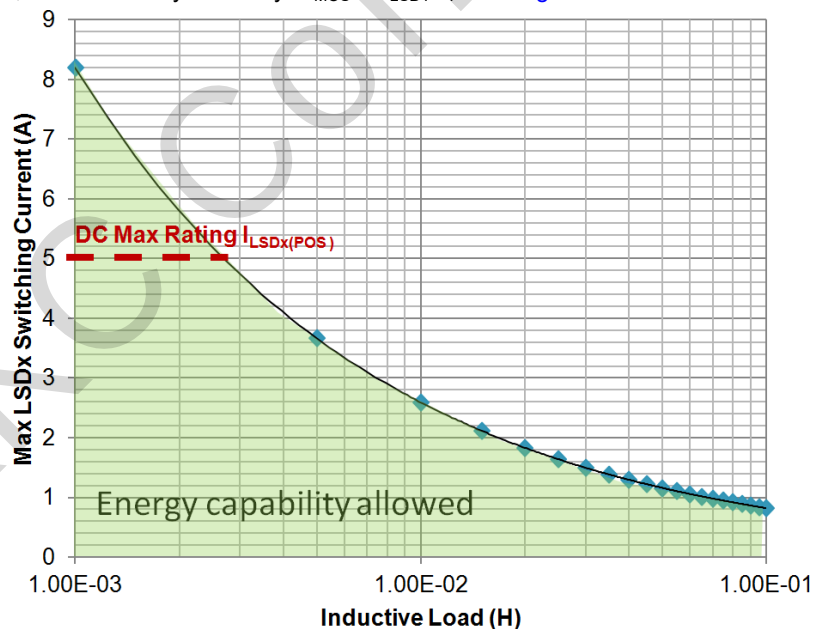


Figure 4-7 LSDx safe operating area for $V_{BAT} = 14\text{ V}$ at $T_A = 125\text{ }^{\circ}\text{C}$ general description

4.6 Wheel speed sensor interface

4.6.1 Supply and protection

This HE0400 provides two protected high-side switches tied to VPWR, which can be switched ON/OFF individually via the SPI. The status of each switch is reported to the MCU via the SPI. The wheel speed conditioning converts a signal given by an active sensor into a signal suitable for a MCU digital input. The input signal of an active sensor is a rectangular signal with variable pulse width and output currents of two levels (7/14 mA) or 3 levels (7/14/28 mA) nominal. The sensor types supported are standard active 2-level wheel speed sensors (7/14 mA), smart active 2-level wheel speed sensors (7/14 mA), and VDA type 3-level wheel speed sensors (7/14/28 mA).

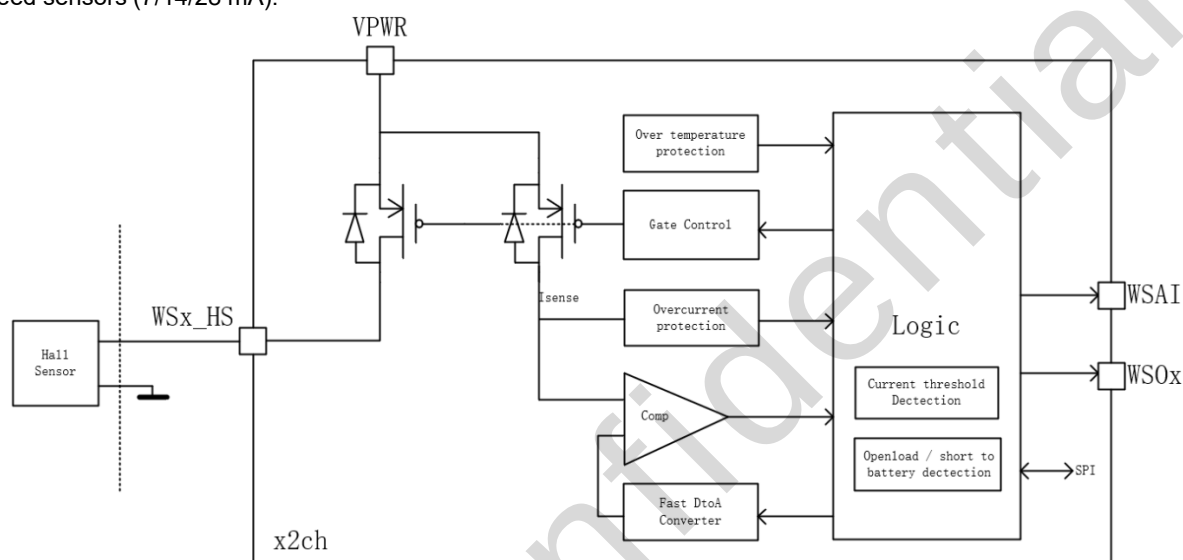


Figure 4-8 Wheel speed sensor interface

The sensor supply consists of a high-side sensor switch WSx_HS per channel. The transistors are automatically enabled after power up. Per the SPI bits, WSx_HS can be switched ON or OFF individually. Each WSx_HS is protected independently.

4.6.1.1 Open load/short to battery

A sensor open load/short to battery condition is detected when the current is below the threshold for the defined filter time and only the fault bit is set (SPI error flag only). This function only works during the ON state.

4.6.1.2 Overcurrent

When the current is above the overcurrent threshold for the defined filter time, the driver is switched off, a SPI fault bit is set and the turn-on SPI command is cleared. The high-side driver can be turned back to a “normal state” by a write 1 to WS_HS_clr_fit, and then turned on by a SPI command. Overcurrent filter time is selectable by a SPI command (See [SPI and data register](#)).

4.6.1.3 Overtemperature

When the temperature is above the overtemperature threshold for the defined filter time, the driver is switched off, a SPI fault bit is set and the turn-on SPI command is cleared. The high-side driver can be turned back to a “normal state” by the temperature returning to a normal state, a SPI write 1 to WS_HS_clrflt, and then a SPI turn on (See [SPI and data register](#)).

4.6.1.4 Overvoltage regulation

When V_{PWR} is higher than the normal voltage, the WSx_HS module supplies a regulated voltage range for the external wheel speed sensor power input (WSx_HS). The ISO pulse on WSx_HS causes WSO toggling and creates an overcurrent or open load fault. To decrease this malfunction, HE0400 WSx_HS implements digital filtering. If the SPI command ISO_det = 1 and ISO_fit = 0, WSOx is fixed/latched during the tiso_fit1 period. Meanwhile, if ISO_det = 1 and ISO_fit = 1, WSOx is fixed/latched during the tiso_fit2 period. The different flags ISO_neg and ISO_pos are reported, whatever the bit value of ISO_det = X. ISO_det = 1 and DF_ws = 0 is a forbidden combination.

Table 4-8 Wheel speed sensor interface matrix

ISO_fit	ISO_det	DF_ws	OCF_ws	Description
1	1	1	1	ISO_fit = 100 us, ISO_det = active, DF_ws = 18 us, OCF_ws = 4xt1 = 72 us, flag reported ISO_pos and neg, if present
0	1	1	1	ISO_fit = 50 us, ISO_det = active, DF_ws = 18 us, OCF_ws = 4xt1 = 72 us, flag reported ISO_pos and neg, if present
1	1	1	0	ISO_fit = 100 us, ISO_det = active, DF_ws = 18 us, OCF_ws = 1xt1 = 18 us, flag reported ISO_pos and neg, if present
0	1	1	0	ISO_fit = 50 us, ISO_det = active, DF_ws = 18 us, OCF_ws = 1xt1 = 18 us, flag reported ISO_pos and neg, if present
1	0	1	1	ISO_fit = 100 us, ISO_det = inactive, DF_ws = 18 us, OCF_ws = 4xt1 = 72 us, flag reported ISO_pos and neg, if present
0	0	1	1	ISO_fit = 50 us, ISO_det = inactive, DF_ws = 18 us, OCF_ws = 4xt1 = 72 us, flag reported ISO_pos and neg, if present
1	0	0	1	ISO_fit = 100 us, ISO_det = inactive, DF_ws = 0 us, OCF_ws = 4xt1 = 72 us, flag reported ISO_pos and neg, if present
	0	0	1	ISO_fit = 50 us, ISO_det = inactive, DF_ws = 0 us, OCF_ws = 4xt1 = 72 us, flag reported ISO_pos and neg, if present
1	0	1	0	ISO_fit = 100 us, ISO_det = inactive, DF_ws = 18 us, OCF_ws = 1xt1 = 18 us, flag reported ISO_pos and neg, if present
0	0	1	0	ISO_fit = 50 us, ISO_det = inactive, DF_ws = 18 us, OCF_ws = 1xt1 = 18 us, flag reported ISO_pos and neg, if present
1	0	0	0	ISO_fit = 100 us, ISO_det = inactive, DF_ws = 0 us, OCF_ws = 1xt1 = 18 us, flag reported ISO_pos and neg, if present
0	0	0	0	ISO_fit = 50 us, ISO_det = inactive, DF_ws = 0 us, OCF_ws = 1xt1 = 18 us, flag reported ISO_pos and neg, if present

Table 4-9 Wheel speed sensor interface electrical characteristics

V_{PWR} = 6.0 V to 20 V, V_{CC5} = 4.75 V to 5.25 V, $DOSV$ = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
WS sensor high-side output						
RON_WS	On resistance	20	27	54	Ω	P_4.9.1
ILEAK_WS	Leakage current, V_{PWR} = 14 V, V_{CC5} = 0, WSx_HS = 0	—	—	2.5	uA	P_4.9.2
t_WS_R t_WS_F	Rise/Fall time - 10% to 90% of WSx_HS, RLOAD = 1.0 kΩ, no capacitor	—	—	5.0	us	P_4.9.3
WS sensor open						
IWS_OPEN	Sensor open detection threshold - same with ISO_pos condition	1.4	2.5	3.5	mA	P_4.9.4

tWS_OPEN	Sensor open detection delay time	—	2xT2	—	us	P_4.9.5
tOP_LSD	ISO positive delay time - guaranteed by design	—	2.0	6.0	us	P_4.9.6

WS sensor overcurrent

WS_OC	Overcurrent shutdown threshold - same with ISO_neg condition	40	60	90	mA	P_4.9.7
tMSK_WS_OC	OC WSx_HS masking time - masking time applies only after off/on transition	—	T1	—	us	P_4.9.8

Table 4-10 Wheel speed sensor interface electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
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WS sensor overcurrent (continued)

tWS_OC1	Overcurrent shutdown filter time	—	T1 4*T1	—	us	P_4.10.1
tWS_OC2	OCF_ws=0 (SPI bit)	—	—	—	us	P_4.10.2
	OCF_ws=1 (SPI bit)	—	—	—	us	P_4.10.2
	ISO negative delay time - guaranteed by design	—	2.0	6.0	us	P_4.10.3

Overtemperature shutdown

OT_WSXHS	Overtemperature detection threshold	180	195	210	°C	P_4.10.4
tOT_WSXHS	Overtemperature detection filter time	—	T1	—	us	P_4.10.5

WS sensor overvoltage

WS_OV	Overvoltage regulation threshold	16	22	24	V	P_4.10.6
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ISO pulse detection

tISO_FLT1	Latched duration of wsox after ISO pulse detection	40	50	60	us	P_4.10.7
tISO_FLT2	ISOflt=0	80	100	120	us	P_4.10.8
	ISOflt=1	—	—	—	us	P_4.10.8

4.6.2 Wheel speed sensor input signal conditioning

The wheel speed information is provided on two dedicated pins (one for each wheel). Each edge of the rectangular signal with a duty cycle of approximation 50% represents a tooth edge in the pole ring. The duty cycle of the standard sensor (Type1) is not modified by the HE0400. A pass-through mode for PWM-encoded sensors (Type2) and pulse encoded sensors (Type3) are selectable via the state of WScfgx<1:0> (SPI command). In those modes, the HE0400 sends decoding data to the MCU.

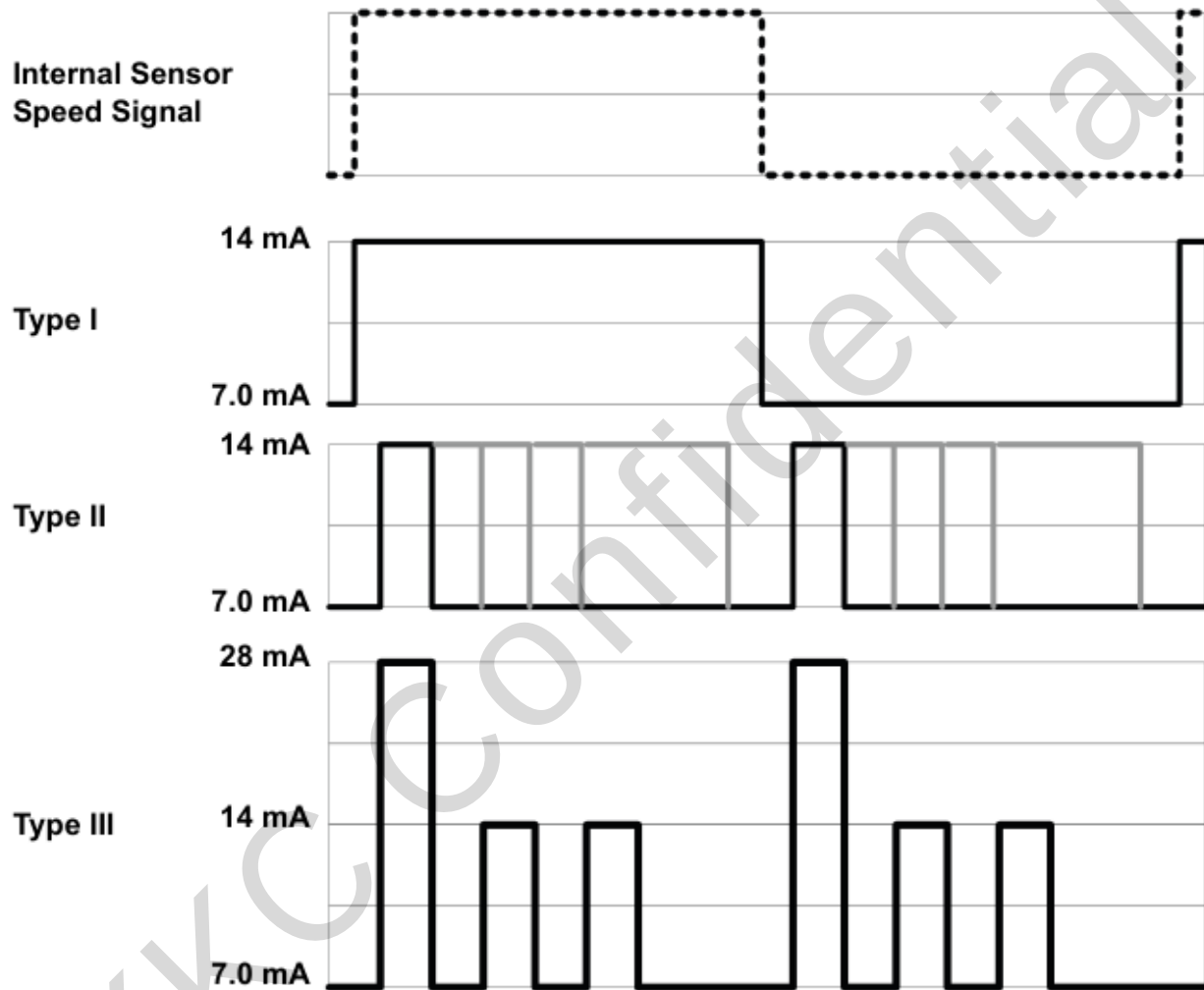


Figure 4-9 Pulses of three types of wheel speed sensors

4.6.2.1 Leakage current

If I_{LOW} goes over the fault tracking level (I_{LEAK_FLT}) for a time period longer than t_{LEAK_FLT} , the related flag WSx_lkg of the SPI register is set high. To solve issues of the WSO/VSO changing state 1.0 s after the motorcycle is stopped ($I_h = 12.9$ mA (typ, time which $wsox$ is transitioned from a 1 to a zero should be used):

- For sensor type I only. (no issue with sensor type II & III)

4.6.2.2 Wheel sensor additional information output

The WSAI pin transfers an additional information pulse (10 mA) to the MCU, which is selected by the SPI command ($WSAI_S0$ and $WSAI_S1$).

SPI command	Wheel sensor type	WSOx	VSO
$WScfgx<1:0>$	"0X"	1	Inverse WSOx
	"10"	2	Inverse WSOx
	"11"	3	Inverse WSOx

WSAI pin follows one of the $WSI10_x$ (10 mA threshold signal):

SPI command	Description
$WSAI_S<1:0>$	"00" WSI10_1 input is selected for WSAI
	"01" WSI10_2 input is selected for WSAI

Default state, after a reset by a certain reset source, is selected WSI10_1 (10 mA threshold)

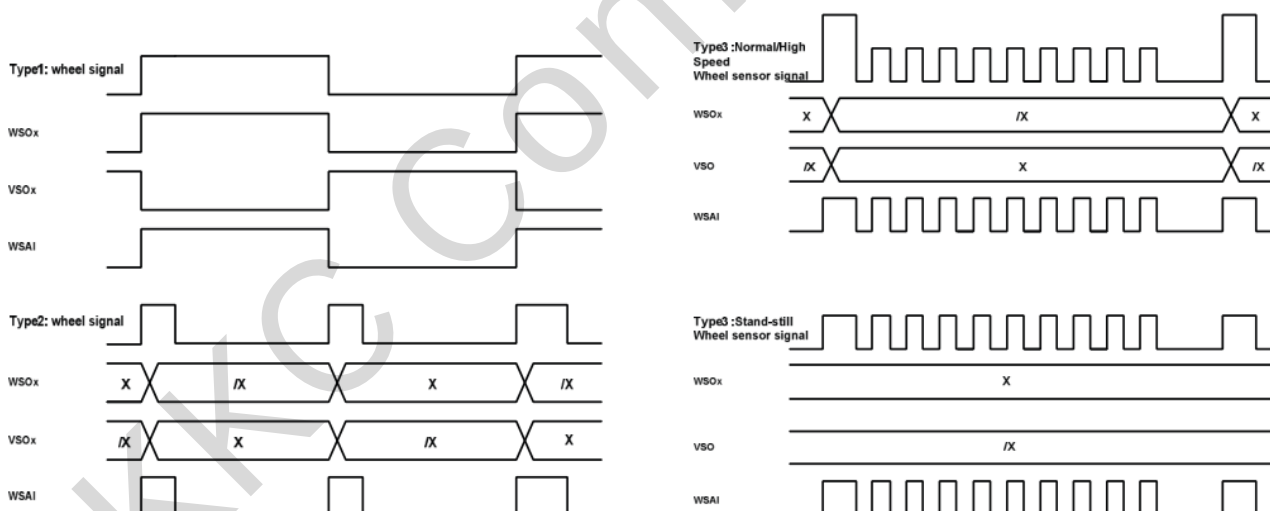


Figure 4-10 Outputs for each wheel speed sensor type

Table 4-11 Wheel speed sensor input signal conditioning electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, $T_J = -40\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$, unless otherwise specified.

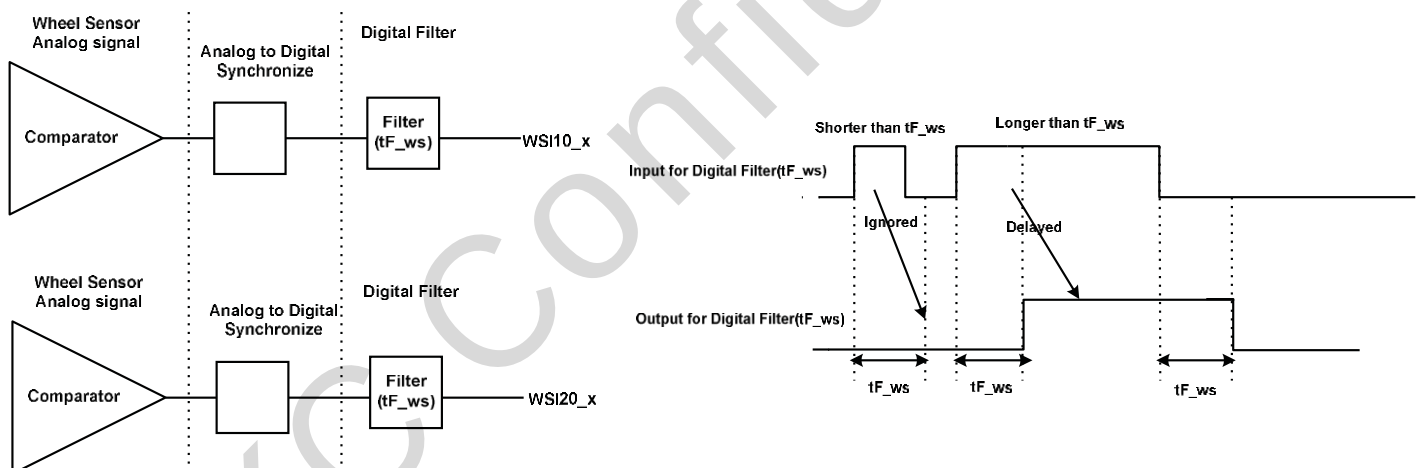
Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Signal current detection ⁽¹⁸⁾						
I_{TH_A}	Input Threshold A - Low current level $I_{LOW} = 7.0\text{ mA}$ ⁽¹⁶⁾	8.75	10.25	11.75	mA	P_4.11.1
I_{TH_B}	Input Threshold B - Low current level $I_{LOW} = 7.0\text{ mA}$ ⁽¹⁶⁾	17	20	23	mA	P_4.11.2
I_{WS_LOW}	Sensor Current for Low Level - Not tested (for information only)	5.9	7.0	8.4	mA	P_4.11.3
I_{LEAK}	Leakage Range - Not tested (for information only)	0.0	—	4.0	mA	P_4.11.4
I_{LOW}	I_{LOW} Range (= $I_{WS_LOW} + I_{LEAK}$)	5.9	7.0	12.4	mA	P_4.11.5
I_{LEAK_FLT}	Fault Leakage Level - measure I_{LOW} ⁽¹⁷⁾	12.4	—	18	mA	P_4.11.8
t_{LEAK_FLT}	Fault Leakage Detection Filter Time - clear read flag SPI	—	T2	—	mA	P_4.11.9

Notes

16. NA.

17. Wheel speed input detection is guaranteed until I_{LOW} go to I_{LEAK_FLT}

18. NA.

**Figure 4-11 Wheel speed sensor interface between analog to digital filter (tF_{WS})**

4.6.3 Wheel speed counter

The wheel speed counter module is composed of two inputs from the current threshold detection, multiplexed

by a 2-bit SPI command (WS_S0 and WS_S1) and one 8-bit counter.

SPI command		Description
WS_S0/WS_S1	"00"	WS01 input is selected for counting
	"01"	WS02 input is selected for counting

When the SPI register "WS_CNT_RST" is set high, the counter is reset (CNT0 to 7 set to "0") during one internal clock cycle, after the rising edge of CSB. When the SPI register "WS_CNT_RST" is set low it has no effect on the counter. When the SPI register "WS_CNT_EN" is set high, the counter starts counting based on signal from the multiplexer output after an internal clock cycle. The counter stops counting when WS_CNT_EN goes to a low state and keeps the last value.

Table 4-12 WS_WS_CNT_RST and WS_CNT_EN

WS_CNT_RST	WS_CNT_EN	Effect on counter
0	0	Counter stopped, keep previous value
0	1	Counter running start with previous value (continue to run if WS_CNT_EN was already 1 before)
1	0	Counter reset at CSB rising edge & stopped at 0 value
1	1	Counter reset at CSB rising edge and start to run started from 0 value

Table 4-13 Wheel speed counter electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
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Wheel speed counter

f _{OP}	Maximum input Frequency - for information only	DC	—	20	kHz	P_4.13.1
t _{F_CNT}	Integrator filter time	14	18	22	us	P_4.13.2

Digital filter for WSI10, WSI20

t _{F_WS}	Rising/falling edge detection filter time - The filter time is selectable by SPI command (DF_ws) ⁽¹⁹⁾	14	18	22	us	P_4.13.3
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Notes

19. Type1, 2, 3; no effect on duty cycle of input (See [Figure 4-11](#)).

4.6.4 Wheel speed information decoding (Type 2, 3)

The HE0400 provides a decoding function for Type 2 and Type 3 wheel speed information. Each duty cycle of the Type 2 sensor and Manchester code of the Type 3 sensor represents wheel direction and some fault mode of each wheel. That information is decoded by the HE0400 and transferred to the MCU by a SPI message. In order to read SPI decoded data, it should write the wheel speed sensor type configuration (WSSxcfg<1:0>) once, even though Type 0 is used. Otherwise, a default condition is transferred.

4.6.4.1 Type 2

Each zero crossing of the magnetic signal triggers an output pulse for the type 2 wheel speed sensor. In addition to the speed signal, the following information is provided by varying the length of the output pulses in [Figure 4-12](#). If a Type 2 or Type 1 is initialized, decoding data when WSSxcfg = 0X is same as one when WSSxcfg = 10. Default condition, all 8 bits are "1", a message go to default after every reading of the SPI.

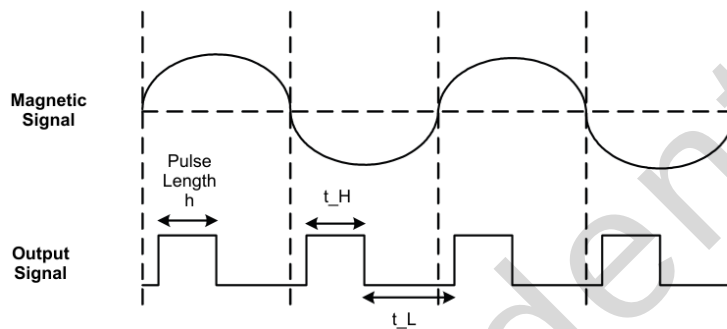


Figure 4-12 Definition t_H and t_L

4.6.4.2 Air-gap warning range = warning

Warning information is issued in the output pulse length, when the magnetic field is below a critical value (e.g. the air gap between the Hall Effect IC and the target wheel exceeds a critical value). The device works with reduced functionality. Warning information is given only in the calibrated mode.

4.6.4.3 Assembly position range = EL

EL information is issued in the output pulse length when the magnetic field is below a predefined value (the air gap between the Hall Effect IC and the target wheel exceeds a predefined value). The device works with full functionality.

4.6.4.4 Direction of rotation right = DR-R

DR-R information is issued in the output pulse length when the target wheel in front of the Hall Effect IC moves from the GND pin to the VCC5 pin.

4.6.4.5 Direction of rotation left = DR-L

DR-L information is issued in the output pulse length when the target wheel in front of the Hall Effect IC moves from the VCC5 pin to the GND pin. At sufficient magnetic field, the direction information is corrected after two pulses during an uncalibrated mode.

4.6.4.6 Magnetic differential

Between each magnetic transition and the rising edge of the corresponding output pulse, the output current is low for $t_{PRE-LOW}$, to allow a reliable internal conveyance. If the magnetic differential field exceeds the assembly position threshold, the output pulse lengths are 90 μ s or 180 μ s respectively, depending on the direction of rotation. When the magnitude of the magnetic differential field is below the assembly position threshold, the output pulse lengths are 360 μ s and 720 μ s respectively, depending on left or right rotation. If the magnitude of the magnetic differential field is below the limit-air gap warning threshold, the output pulse length is 45 μ s. The warning output is dominant. This means it is close to the limit-air gap. The direction and the assembly position information are disabled. If no magnetic differential signal is detected for a time longer than the stand still period t_{STOP} , it outputs a stop pulse.

Table 4-14 Type 2 sensor pulse signal timing (for information, not tested)

Parameter	Min.	Typ.	Max.	Unit
Signal: start bit low	38	45	52	us
Signal: LR = air gap limit	38	45	52	us
Signal: DRL = Rotation direction left	75	90	104	us
Signal: DRR = Rotation direction right	153	180	207	us
Signal: DRL&EL = Rotation direction left & bad band gap	306	360	414	us
Signal: DRR&EL = Rotation direction right & bad band gap	616	720	828	us
Signal: standing wheel	1232	1440	1656	us
Time detect standing wheel	590	737	848	ms

Speed signal (internal sensor)

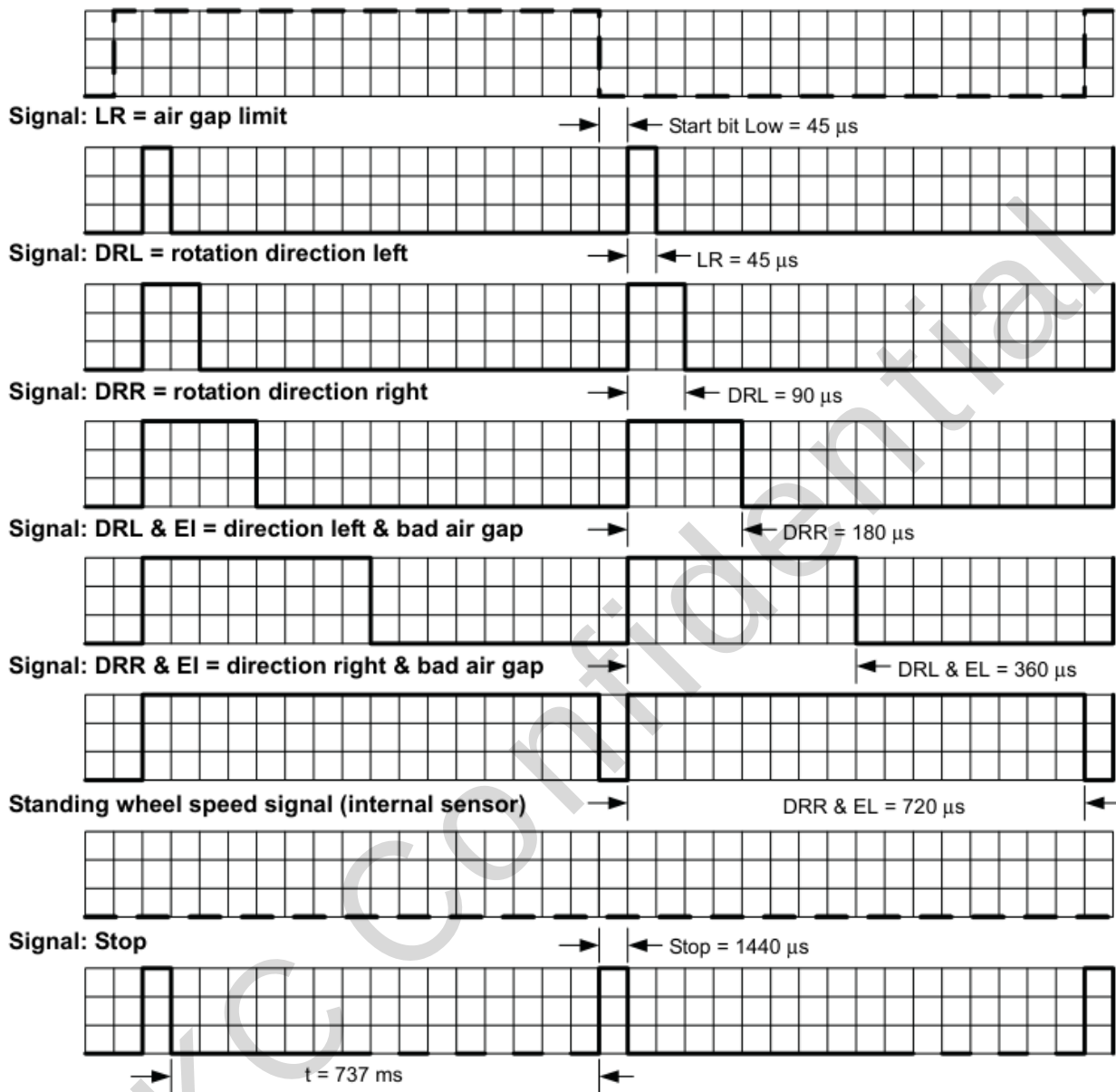


Figure 4-13 Type 2 sensor pulse signal timing

Decoding information transfers to the MCU by an 8-bit stream of the SPI. Sensor testing and monitoring information has a defined data bit address for each sensor input. If t_L is shorter than t_{th8} , the information of the t_H pulse length cannot be valid data.

Table 4-15 Type 2 sensor information

Information	Pulse length	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Signal Failure	$t_H < t_{TH1}$, $t_H < 32 \mu s$	0	0	0	0	0	0	0	0
LR	$t_{TH1} < t_H < t_{TH2}$, $32 < t_H < 64 \mu s$	1	0	0	0	0	0	0	0
DR-L	$t_{TH2} < t_H < t_{TH3}$, $64 < t_H < 128 \mu s$	0	1	0	0	0	0	0	0
DR-R	$t_{TH3} < t_H < t_{TH4}$, $128 < t_H < 256 \mu s$	0	0	1	0	0	0	0	0
DR-L & EL	$t_{TH4} < t_H < t_{TH5}$, $256 < t_H < 512 \mu s$	0	0	0	1	0	0	0	0
DR-R & EL	$t_{TH5} < t_H < t_{TH6}$, $512 < t_H < 1024 \mu s$	0	0	0	0	1	0	0	0
Stop	$t_{TH6} < t_H < t_{TH7}$, $1024 < t_H < 2016 \mu s$	0	0	0	0	0	1	0	0
Sensor Failure	$t_H > t_{TH7}$, $t_H > 2016 \mu s$	0	0	0	0	0	0	1	0
Information Invalid	$t_H < t_{TH8}$, $t_L < 64 \mu s$	0	0	0	0	0	0	0	1

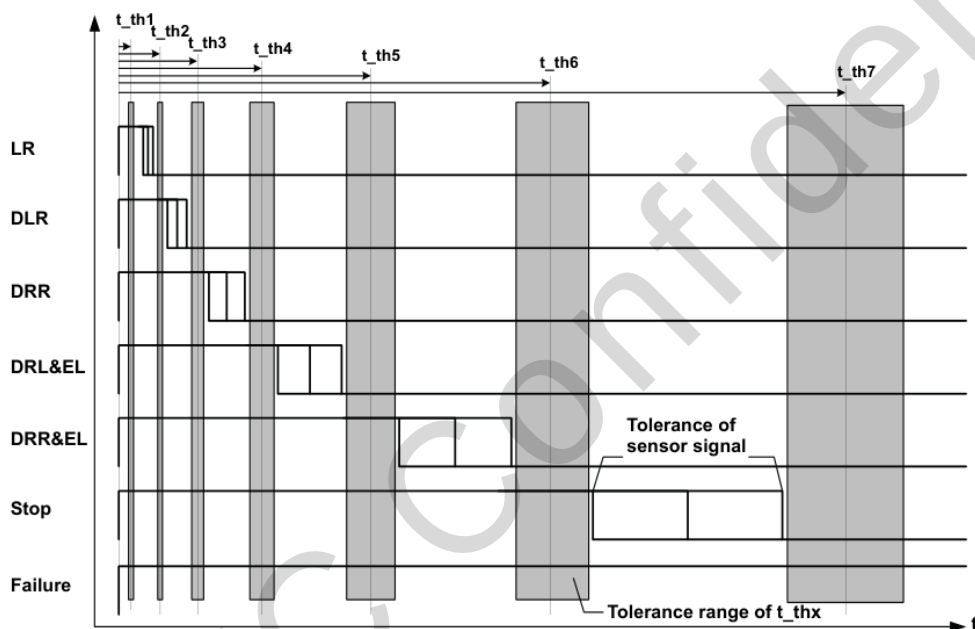


Figure 4-14 Decoding threshold time

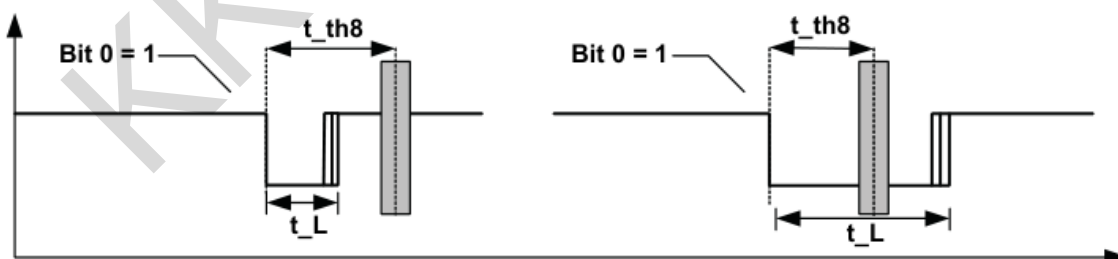


Figure 4-15 Signal monitoring - information invalid

Table 4-16 Signal monitoring electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Timing						
t_{TH1}	Threshold level 1	27	32	37	us	P_4.16.1
t_{TH2}	Threshold level 2	53	64	74	us	P_4.16.2
t_{TH3}	Threshold level 3	105	128	152	us	P_4.16.3
t_{TH4}	Threshold level 4	208	256	305	us	P_4.16.4
t_{TH5}	Threshold level 5	415	512	615	us	P_4.16.5
t_{TH6}	Threshold level 6	829	1024	1231	us	P_4.16.6
t_{TH7}	Threshold level 7	1657	2016	2420	us	P_4.16.7
t_{TH8}	Threshold level 8	53	64	74	us	P_4.16.8

Table 4-17 The SPI message of type 2 default information

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	WS1_lkg	(WSI1)data					Stop	Fail	Invalid	0	0	0	0	0
X	X	X	1	1	1	1	1	1	1	1	X	X	X	X	X

4.6.4.7 Type 3

For the Type 3, wheel speed sensor, the speeds are transferred as the cyclic pulses and the additional information is transferred between the speed pulses, with the help of a serial data log.

The speed pulse is introduced by an initial bit, which sets the defined current level to a I_{low} beforehand. The speed pulse has the current I_{high} level and does not necessarily have to be located at the position of the terminal change. It is transferred as a rectangular pulse with a I_{high} level over the entire pulse duration, and is not subject to Manchester coding as the other log bits.

The bits 0 - 8 in Figure 4-16 are designed as the data log, and their purpose is to transfer the additional information via the speed pulse. The data log should be implemented with a Manchester code, whereby binary values are output by flanks, which are located in the middle in the corresponding time interval. The number of the log bits, when at a standstill and at normal speed, is defined on 9 (bit 0 to 8). This means more or less than 9 is not permitted. Unused bits should be output as default values.

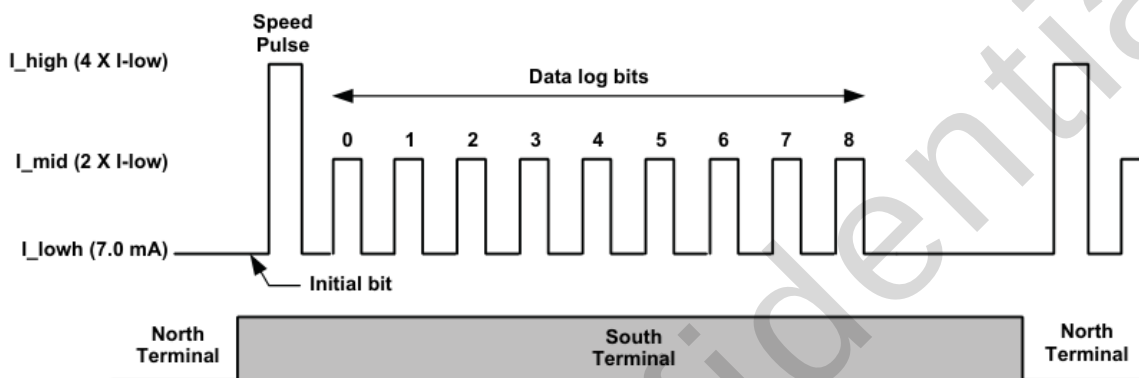


Figure 4-16 Definition of speed pulse and data log bits

4.6.4.8 Normal speed condition

This is commenced with an initial bit, which occupies I_{low} for $t_p/2$, followed by the speed pulse with the level I_{high} for the duration of t_p , then followed by a level I_{low} for a duration of $t_p/2$. This is necessary, so that before the first additional bit (bit 0), the current level of I_{low} is always achieved. Otherwise, in the case of $bit_0 = "1"$, the current level would jump directly from a I_{high} to a I_{mid} , which is not desirable. The data log follows. In the normal speed range, all data bits can be transferred. The subsequent speed pulse of the next log cycle does not cut off the serial data log.

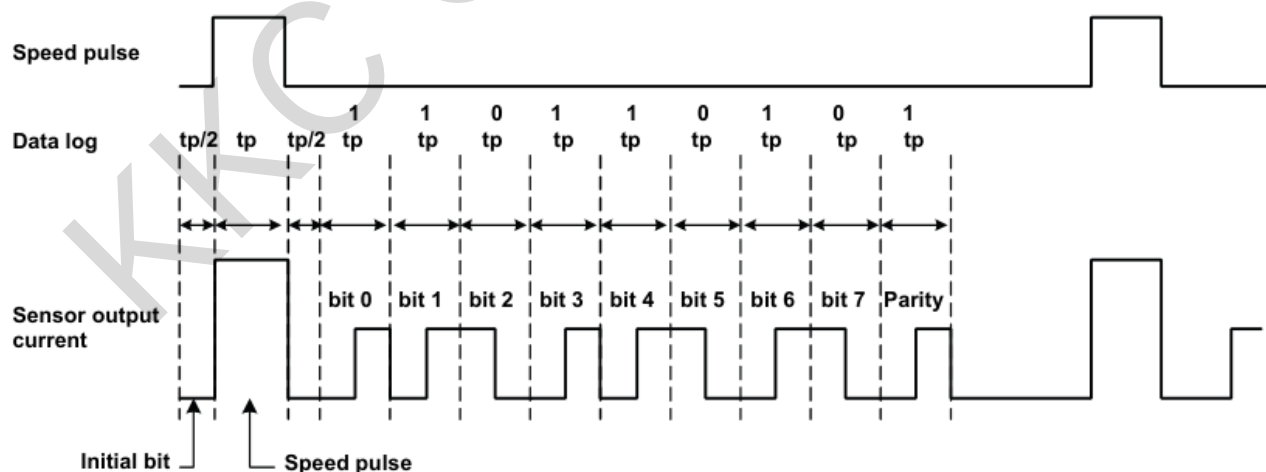


Figure 4-17 Signal monitoring: normal speed

4.6.4.9 High speed condition

The initial bit I_{low} always precedes the speed pulse for at least $t_P/2$. The serial data log is shortened at high speeds, because the time to the next speed pulse is shorter than the log cycle. The data bits at the end are therefore "cut off". In each speed range, the maximum possible number of bits of additional information should be transferred. If there is no edge during the inter-frame timing of high speed (t_{MC_IF}), all data bits of the current frame is invalid and the error code flag goes high.

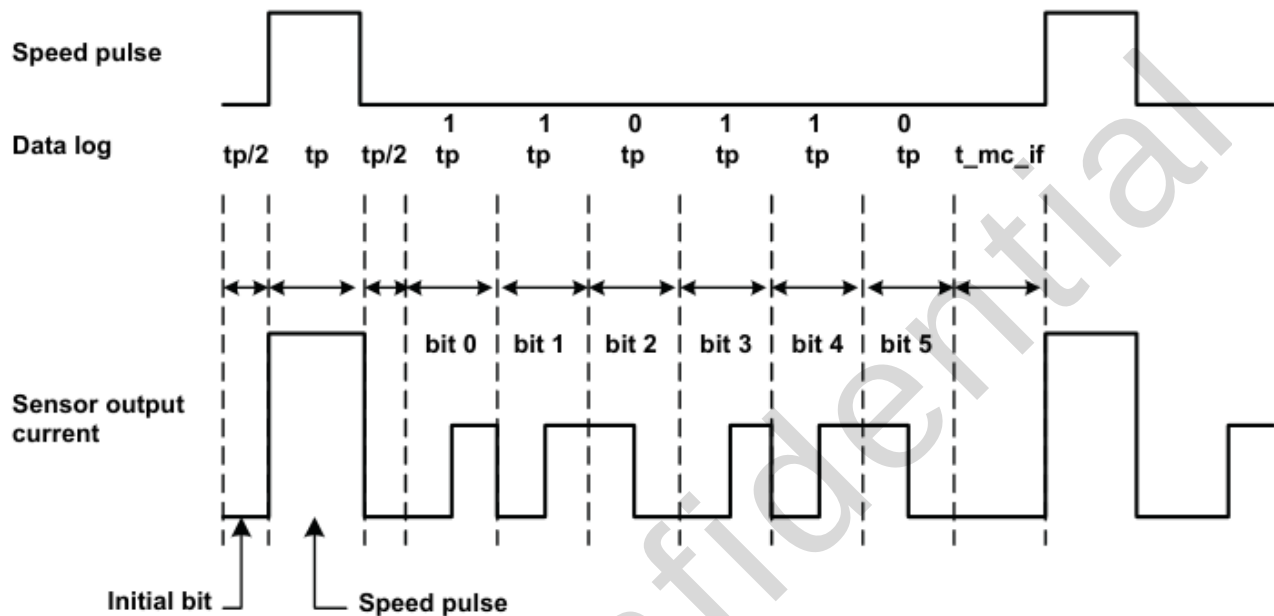


Figure 4-18 Signal monitoring: high speed

Table 4-18 Availability of the data bits depending on encoder frequency

Signal frequency	Time between speed pulse	Available data bits
< 1,800 Hz	555 μ s	bit 0 to bit 8
< 2,000 Hz	500 μ s	bit 0 to bit 7
< 2,200 Hz	455 μ s	bit 0 to bit 6
< 2,400 Hz	417 μ s	bit 0 to bit 5
< 2,800 Hz	357 μ s	bit 0 to bit 4
< 3,200 Hz	313 μ s	bit 0 to bit 3
< 4,000 Hz	250 μ s	bit 0 to bit 2
< 5,000 Hz	200 μ s	bit 0 to bit 1

4.6.4.10 Stand still condition

The serial data log is still sent during standstill to make additional information available, even under these conditions. This is commenced with an initial bit which occupies I_{low} for $t_p/2$. It is followed by the speed replacement pulse for the duration t_p , which in contrast to the speed pulse, occupies the current level I_{mid} . This ensures the wheel speed sensor (transmitter) and the HE0400 (receiver) can be synchronized with one another, even if no speed pulse has been transmitted. A level I_{low} follows the speed replacement pulse for a period of $t_p/2$. Finally, the data log bits follow. If the sensor does not detect an increment within t_{STOP} , the data log is transmitted again after a period of t_{STOP} . This cyclical procedure is called the “standstill log”.

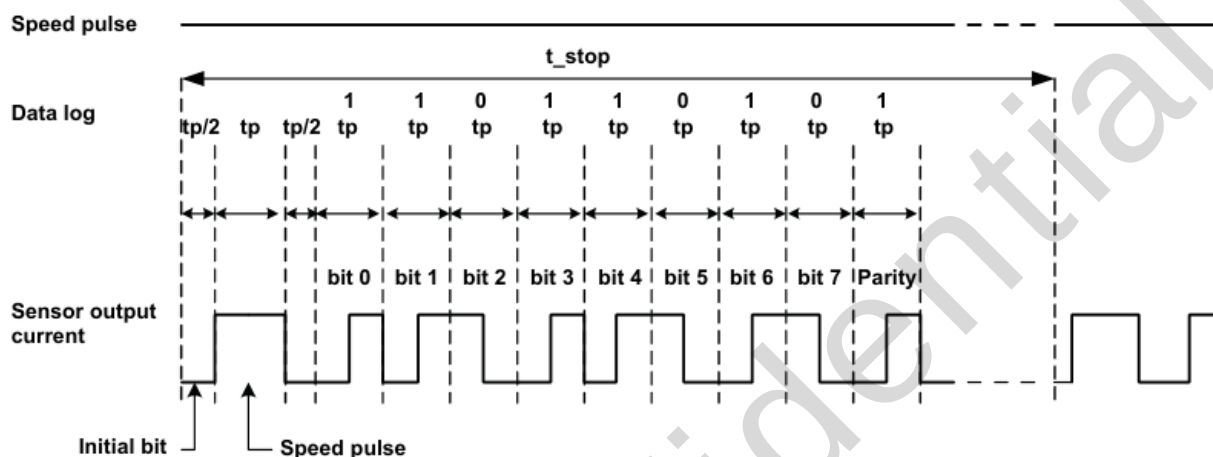


Figure 4-19 Signal monitoring: stand still

Manchester decoding result transfers a 9-bit stream of the SPI to the MCU. The following 3-bits indicate the number of invalid bits in high speed mode. Code error means a Manchester decoding error where no edge is detected in the detection window. During a code error, the Manchester decoding result keeps the previous data and a code error bit goes to “1”. The code error flag is reset by a SPI read. The default condition indicates all 12 bits are “1”, a message goes to a default after every reading of the SPI.

Table 4-19 SPI message for type 3 default information

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	WS1_lkg	(WS11) Manchester decoding result									code error	Number of Invalid Bits		
X	X	X	1	1	1	1	1	1	1	1	1	0	1	1	1

Table 4-20 Start and stop condition for manchester decoding

Condition	Decoding start	Decoding stop
Normal speed	10 mA pulse with t_p (AND) 20 mA pulse with t_p	Receive 9 bits (OR) Rising edge of 20 mA pulse
High speed	10 mA pulse with t_p (AND) 20 mA pulse with t_p	Rising edge of 20 mA pulse
Stand still	10 mA pulse with t_p	Receive 9 bits

Decoding start can be restarted after the release of the decoding stop condition. After receiving the start point condition, the HE0400 begins decoding. It is possible for wrong decode information due to a noise pulse during the Manchester coding period. If the pulse width of the start pulse is longer than t_{P_MAX} or shorter than t_{P_MIN} , decoding should not start. If no message is received before “ t_{STOP} ”, the HE0400 detects a timeout error.

Table 4-21 Stand still condition electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, $T_J = -40\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Timing						
t_P	Pulse width for speed pulse Pulse width for data pulse (20)	-20% -20%	50 50	+20% +20%	us	
$t_{P/2}$	Bit time for data bits	-20%	25	+20%	us	
t_{STOP}	Timeout error without data (21)	-20%	240	+20%	ms	
t_{P_MIN}	Short start pulse detection (22)	30	35	40	us	
t_{P_MAX}	Long start pulse detection (22)	60	75	90	us	
t_{MC_IF}	Max. inter-frame timing for high speed condition	2.0	2.3	2.7	ms	

Notes

20. Not tested (Only for information)

21. Start timer at validation of 9-bit data message. Application note: standard stand still period is 150 ms $\pm$ 30% only for type 3

22. No start of decoding

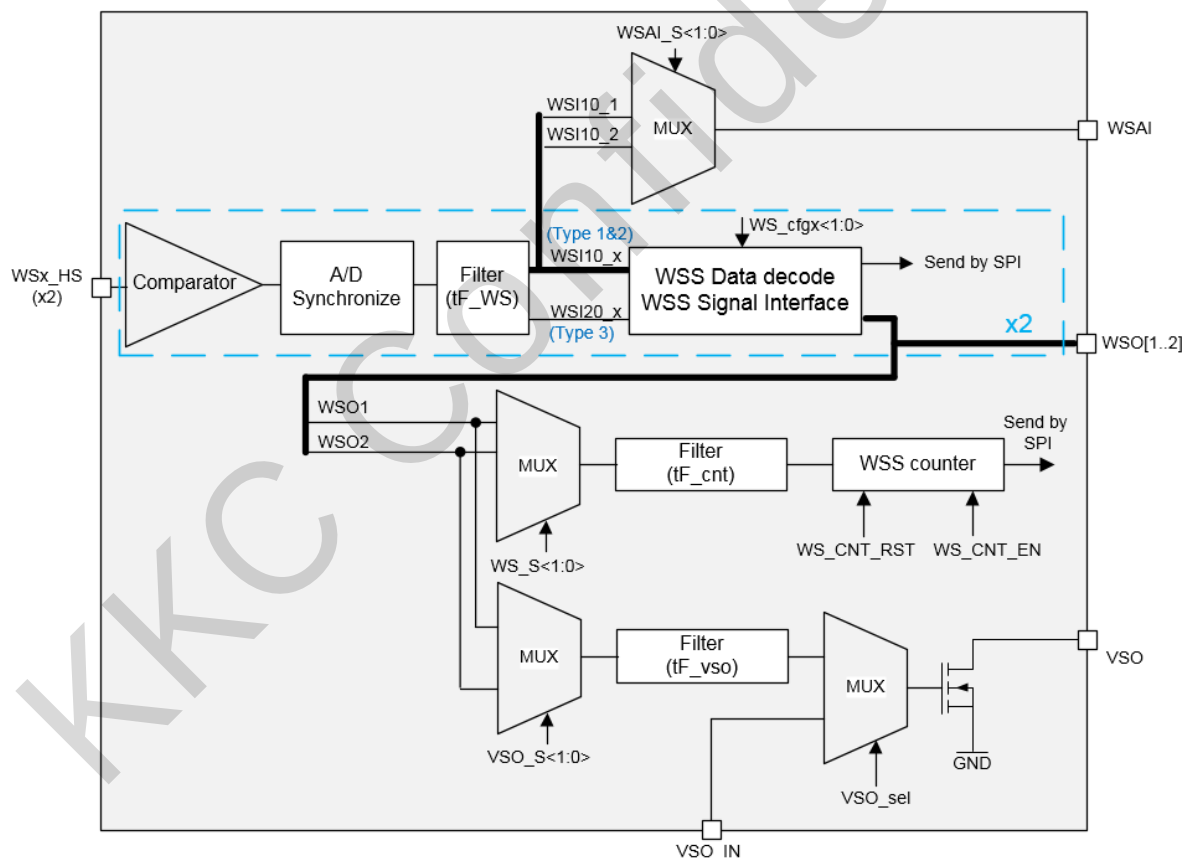


Figure 4-20 Diagram of wheel speed sensor interface

4.7 High-voltage outputs VSO

4.7.1 Function description

VSO sends the vehicle speed output signal from WSOx or VSO_IN (external pin), and is selected by the SPI command VSO_sel.

SPI Command		Description
VSO_sel	0	VSO is controlled by WSOx
	1	VSO is controlled by VSO_IN

The pre-driver applies the necessary voltage on the output transistor gate to minimize the on-resistance of the output switch.

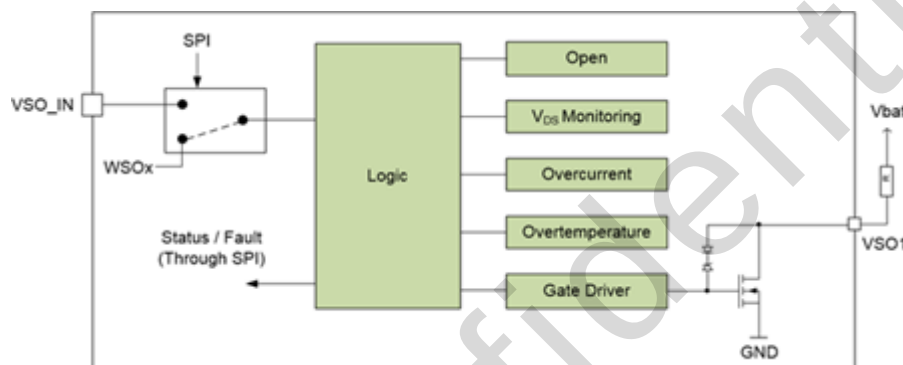


Figure 4-21 VSO driver

4.7.2 Fault detection

4.7.2.1 Open load

An open condition is detected when the VSO output is below the threshold for the defined filter time. The fault bit is set (SPI error flag only). This function is only operating during the off state.

4.7.2.2 V_{DS} state monitoring

V_{DS} state monitoring gives real time state of the LSD drain voltage vs. OP_vso voltage. This signal is filtered and sent through the SPI. If the V_{DS} voltage is higher than OP_vso with a filter time (T1), vds_vso is set to "1".

4.7.2.3 Overcurrent

When the current is above the overcurrent threshold for the defined filter time, the driver is switched off and a SPI fault bit is set. The driver can be turned back to the "normal state" by writing 1 to VSOx_clr_ft, turning it on automatically. VSO follows WSOx or VSO_IN by the previous SPI command (vso_sel).

4.7.2.4 Overtemperature

When the temperature is above the overtemperature threshold for the defined filter time, the driver is switched off and a SPI fault bit is set. The driver can be turned back to the "normal state" when the temperature returns to the normal state and writing a 1 to VSOx_clr_ft, turning it on automatically. VSO follows WSOx or VSO_IN by the previous SPI command (vso_sel).

Table 4-22 Fault detection electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Power output VSO						
R _{ON_VSO}	On Resistance T _J = 150 °C, 6.0 V ≤ V _{PWR} ≤ 20 V	—	8.0	14	Ω	P_4.22.1
I _{LEAK_VSO}	Drain Leakage Current V _{PWR} = 0, V _{CC5} = 0, VSO = 14 V	—	—	10	μA	P_4.22.2
	DC Current Capability	—	—	20	mA	P_4.22.3
V _{BVDSS_VSO}	BVDSS Voltage	40	—	—	V	P_4.22.4
I _{NEG_VSO}	Maximum Negative Current for 5.0 ms without being destroyed	100	—	—	mA	P_4.22.5
Timing						
t _{F_VSO}	Rising/falling Edge Detection Filter Time Type1, 2, 3 - no effect on input duty cycle	14	18	22	μs	P_4.22.6
f _{VSO}	Transmission Frequency - R _L = 1.0 kΩ, not tested	—	—	10	kHz	P_4.22.7
t _{D_ON_VSO} / t _{D_OFF_VSO}	Turn On/off Delay Time (23)	—	—	1.0	μs	P_4.22.8
Open load detection						
OP _{VSO}	Open Load Detection Threshold - also used for VDS monitoring	1.6	2.0	2.4	V	P_4.22.9
top _{VSO}	Open Load Detection Filter Time	—	T2	—	μs	P_4.22.10
VDS state monitoring						
t _{VDS_VSO}	VDS State Filter Time (24)	—	T1	—	μs	P_4.22.11
Overcurrent shutdown						
OC _{VSO}	Overcurrent Shutdown Threshold Current	50	—	150	mA	P_4.22.12
toc _{VSO}	Overcurrent Shutdown Filter Time	—	T1	—	μs	P_4.22.13
Overtemperature shutdown						
OT _{VSO}	Overtemperature Detection Threshold	180	195	210	°C	P_4.22.14
tot _{VSO}	Overtemperature Detection Filter Time	—	T1	—	μs	P_4.22.15

Notes

23. From digital signal input to 50%, output R_L = 1.0 kΩ, V_{PWR} = 14 V, no capacitor

24. Used open load detection comparator rise & fall edge filter time

4.8 K-Line interface

4.8.1 Function description

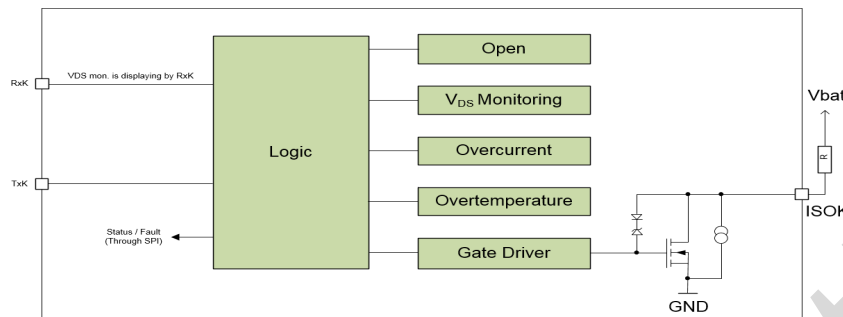


Figure 4-22 K-Line interface

The K-line module is a serial link bus interface device designed to provide bi-direction half-duplex communication interfacing in automotive diagnostic applications. It should be designed to interface between the vehicles via the special ISO K-line, and meet the "Diagnostic Systems ISO9141" specification. The device's K-line bus driver's output is fully protected against a bus short.

- Interfaces directly to standard CMOS Microprocessors
- ISO K Line pin is protected against shorts to battery
- ISO-K line can be driven with up to 10 nF of parasitic capacitance

The K-line module transforms 5.0 V or 3.3 V MCU logic signals to battery level logic signals and vice versa. This serial link interface module, operating in a typical automotive diagnostic application, operates at bit rates up to 10.4 kbps/s, with propagation delay less than 2.0 μ s, and fall time less than 2.0 μ s. It should remain fully functional over a battery voltage range of 6.0 V to 20 V. The RxK and TxK control inputs are compatible with standard 5.0 V or 3.3 V CMOS circuitry.

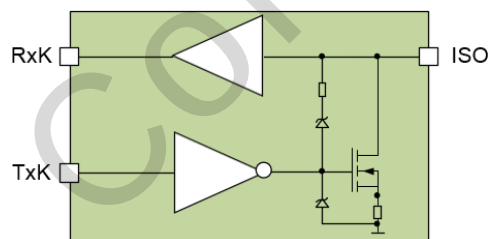


Figure 4-23 K-Line application

It is possible to use the K-LINE module in another application. On this example, we can use this function as a second Vehicle Speed Output by connecting a WSOx externally to the TxK pin. The ISO Input/Output becomes a Vehicle Speed Output 2 (VSO2 here).

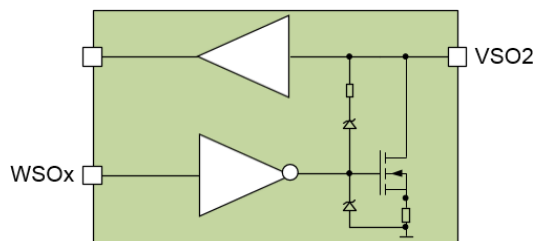


Figure 4-24 VSO2 application

4.8.2 Fault detection

4.8.2.1 Overcurrent

When the current is above the overcurrent threshold OC_{ISOK} for the defined filter time t_{OC_ISOK} , the driver is switched off and a SPI fault bit $isok_OC$ is set. The driver can be turned back to a "normal state" by a SPI write 1 to $ISOK_clr_flt$, followed by a TxK.

4.8.2.2 Overtemperature

When the temperature is above the overtemperature threshold OT_{ISOK} for the defined filter time t_{OT_ISOK} , the driver is switched off and a SPI fault bit $isok_OT$ is set. The driver can be turned back to a "normal state" when temperature returns to a normal state, followed by a SPI write 1 to $ISOK_clr_flt$ and then a TxK.

Table 4-23 K-Line electrical characteristics

$V_{PWR} = 6.0\text{ V to }20\text{ V}$, $V_{CC5} = 4.75\text{ V to }5.25\text{ V}$, $DOSV = 3.13\text{ V to }5.25\text{ V}$, $T_J = -40\text{ °C to }+150\text{ °C}$, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
ISOK						
V_{IL_ISO}	ISOK Input Voltage Threshold	$0.4 \times V_{PWR}$	—	$0.7 \times V_{PWR}$	V	P_4.23.1
R_{ON_ISOK}	On Resistance $T_J = 150\text{ °C}$, $6.0\text{ V} \leq V_{PWR} \leq 20\text{ V}$	—	8.0	14	Ω	P_4.23.2
	DC Current Capability	—	—	40	mA	P_4.23.3
Timing						
t_{PD_ISO}	ISO Propagation Delay TxK High to TxK Low & TxK Low to TxK High (25)	—	—	1.0	μs	P_4.23.4
t_{PD_RX}	RxK Propagation Delay ISOK High to ISOK Low & ISOK Low to ISOK High (26)	—	—	1.0	μs	P_4.23.5
Overcurrent shutdown						
OC_{ISOK}	Overcurrent Shutdown Threshold Current	50	100	150	mA	P_4.23.6
t_{OC_ISOK}	Overcurrent Shutdown Filter Time	—	T1	—	μs	P_4.23.7
Overtemperature shutdown						
OT_{ISOK}	Overtemperature Detection Threshold	180	195	210	$^{\circ}\text{C}$	P_4.23.8
t_{OT_ISOK}	Overtemperature Detection Filter Time	—	T1	—	μs	P_4.23.9

Notes

25. From TxK to ISO $50\% V_{PWR} = 14\text{ V}$, $R_{ISO} = 510\text{ }\Omega$, no capacitor

26. From ISO to RxK $50\% V_{PWR} = 14\text{ V}$, $R_{ISO} = 510\text{ }\Omega$, no capacitor

4.9 Warning lamp driver

4.9.1 Power output stages

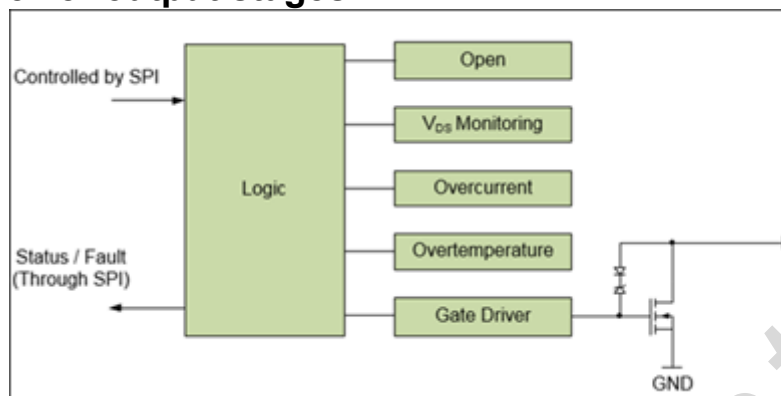


Figure 4-25 Warning lamp driver

The warning lamp driver consists of DMOS-power transistors with an open drain output. The warning lamp driver is controlled by a SPI command (bit"00" message 5). First the bit"02" WLD_en of message 3 should be set to allow the functionality, then the WLD can be turn- on/off by the bit "0" of message 5. The warning lamp driver is composed of an output transistor, a pre-driver circuit, and diagnostic circuitry. The pre-driver applies the necessary voltage on the output transistor gate to minimize On-resistance of the output switch. To avoid a leakage current path, the WLD has no sink current.

Table 4-24 Warning lamp driver electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, $T_J = -40\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Power output WLD						
RON_WLD	On Resistance $T_J = 150\text{ }^{\circ}\text{C}$, $6.0\text{ V} \leq V_{PWR} \leq 20\text{ V}$	—	8.0	14	Ω	P_4.24.1
	DC Current Capability	—	—	20	mA	P_4.24.2
I _{LEAK_WLD}	Drain Leakage Current (27)	—	—	2.0	μA	P_4.24.3
V _{BVDSS_WLD}	BVDSS Voltage	40	—	—	V	P_4.24.4
I _{NEG_WLD}	Maximum Negative Current for 5.0 ms without being destroyed	100	—	—	mA	P_4.24.5
Timing						
t _{D_ON_WLD} / t _{D_OFF_WLD}	Turn on/off Delay Time for WLD (28)	—	—	1.0	μs	P_4.24.6

Notes

27. $V_{PWR} = 0\text{ V}$, $V_{CC5} = 0$, $WLD = 14\text{ V}$, no sink current

28. From digital signal to 50% (turn ON) or 50% (turn OFF). $R_L = 1.0\text{ K}\Omega$, $V_{PWR} = 14\text{ V}$, no capacitor

4.9.2 Fault detection

4.9.2.1 Open load

An open condition is detected when the WLD output is below the threshold OP_{WLD} for the defined filter time t_{OP_WLD} . The fault bit is set wld_OP (SPI error flag only). This function is only operating during the off state.

4.9.2.2 VDS state monitoring

The V_{DS} state monitoring gives the real time state of WLD drain voltage vs OP_{WLD} voltage. This signal is filtered and sent through the SPI vds_wld bit. If the V_{DS} voltage is higher than OP_{WLD} with a filter time ($T1$), vds_wld is set to "1".

4.9.2.3 Overcurrent

When the current is above the overcurrent threshold OP_{WLD} for the defined filter time t_{OC_wld} , the driver is switched off, a SPI fault bit wld_OC is set, and the turn-on SPI command is cleared. The driver can be turned back to a "normal state" by a SPI write "1" to "WLD_clr_flg", then turned on by the SPI command (WLD_on).

4.9.2.4 Overtemperature

When the temperature is above the overtemperature threshold OT_{WLD} for the defined filter time OT_{WLD} , the driver is switched off, a SPI fault bit wld_OT is set, and the turn-on SPI command is cleared. The driver can be turned back to a "normal state" by the temperature returning to a normal state. A SPI write "1" to "WLD_clr_flg", then a turn on by the SPI command (WLD_on).

Table 4-25 Warning lamp driver electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Overcurrent shutdown						
OC_{WLD}	Overcurrent shutdown threshold current for WLD (29)	50	100	150	mA	P_4.25.1
t_{OC_WLD}	Overcurrent shutdown filter time	—	T1	—	μs	P_4.25.2
Open load detection						
OP_{WLD}	Open load detection threshold - also used for V_{DS} monitoring	1.6	2.0	2.4	V	P_4.25.3
t_{OP_WLD}	Open load detection filter time	—	T2	—	μs	P_4.25.4
VDS monitoring						
t_{VDS_WLD}	V_{DS} state filter time (30)	—	T1	—	μs	P_4.25.5
Overtemperature shutdown						
OT_{WLD}	Overtemperature detection threshold	180	195	210	°C	P_4.25.6
t_{OT_WLD}	Overtemperature detection filter time	—	T1	—	μs	P_4.25.7

Notes

29.

30. Used open load detection comparator rise & fall edge filter time

4.10 Monitoring module

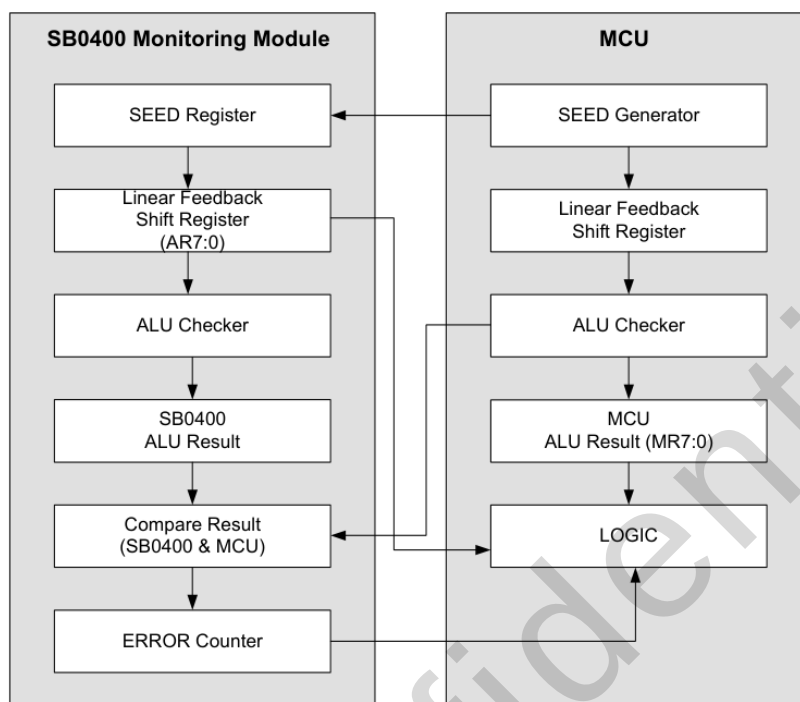


Figure 4-26 Block diagram of HE0400 monitoring module and MCU

The monitoring module in HE0400 works independently from the MCU functionality. The SEED is an 8-bit word which initializes the monitoring module and is transferred by the SPI. The MCU generates the SEED, and should fetch and send correct calculation results (MR7:0) to the HE0400 monitoring module within a defined time window. The HE0400 monitoring module confirms the result is sent and correct in the time window. Linear Feedback Shift Register results of HE0400 monitoring module are transferred to the MCU by the SPI. The monitoring module also calculates the expected correct result, which is compared to the actual result from MCU.

The result from MCU is an 8-bit MR. The 8 bits are sent to the monitoring module via the SPI interface. The monitoring cycle time starts by a write of MR, with the next MR written within in a fixed time window. A new cycle time is started automatically by a write of MR.

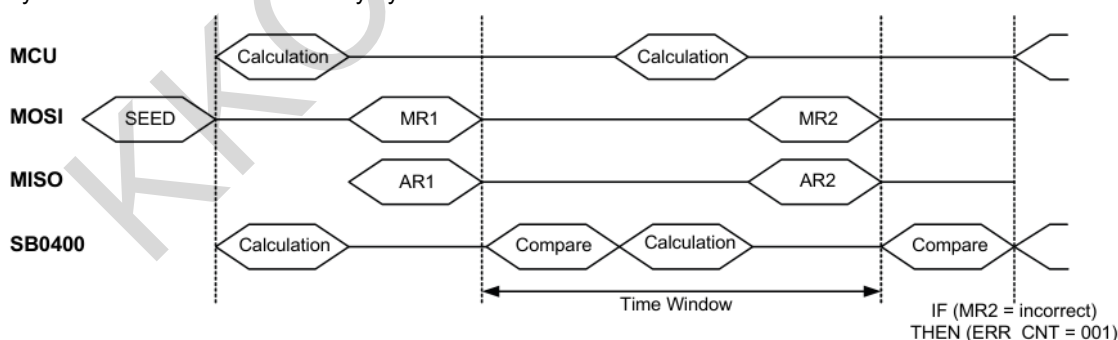


Figure 4-27 Timing diagram of HE0400 monitoring module and MCU

ERR_CNT is a 3-bit counter. An incorrect result leads to incrementing the ERR_CNT by one, and a correct result leads to decrementing by one. The ERR_CNT 3-bit can be read by the SPI interface.

4.10.1 ERR_CNT behavior

Reset with (RSTB pin = "Low")

IF (ERR_CNT $\geq$ 101) THEN (RSTB pin = "Low") AND (MON_CNT reset)

IF (ERR_CNT $\leq$ 100) AND (MR = incorrect) THEN

(ERR_CNT = ERR_CNT + 1) IF (ERR_CNT = 000) AND

(MR = correct) THEN (ERR_CNT = ERR_CNT)

IF (001 $\leq$ MON_CNT $\leq$ 100) AND (MR = correct) THEN (ERR_CNT = ERR_CNT - 1)

The HE0400 monitors the time window of the SPI message #18 without writing to the SEED. The time window (t_{wd}) counter starts with the RSTB pin rising edge. The time window (t_{wd}) counter is reset with the SPI message #18 (with valid parity bit) and restart. If the SPI message #18 (with valid parity bit) is not transferred from the MCU before the time window (t_{wd}) end period, the RSTB pin goes to a LOW state for the duration time of t_{RSTB_REC} and the RST_wd flag is set "High". When RSTB is at a low state (internal, external), the time window (t_{wd}) counter is reset to zero.

4.10.2 Linear feedback shift register (LFSR)

Both the HE0400 monitoring module and the MCU have LFSR for a pseudo-random number generator of ALU checker inputs. LFSR works in parallel with the HE0400 and MCU. LFSR is initialized by the SPI with a SEED 8-bit, then each MR write command generates a new pseudo-random number. FF hex-value cannot be used for the SEED.

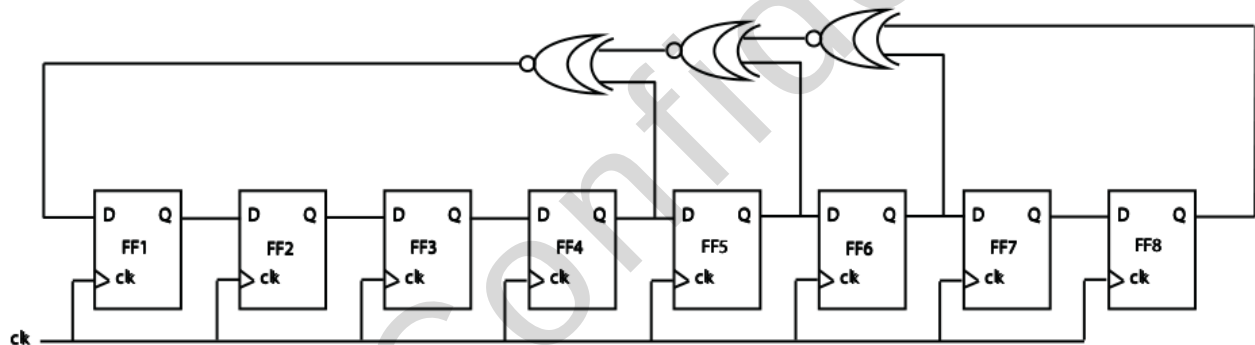


Figure 4-28 Diagram of linear feedback shift register (LFSR)

4.10.3 ALU checker

Both the HE0400 monitoring module and the MCU have an ALU checker. The ALU checker work in parallel with the HE0400 and MCU. The 8-bit input of the ALU checker is the 8-bit output of LFSR. The ALU checker proceeds on five sequential calculations.

- Multiplier pseudo-random value by fix value 4
- Adder output multiplier by fix value 6
- Subtract previous value with fix value 4
- Inverting previous value: bitwise complement
- Divider previous value by fix value 4

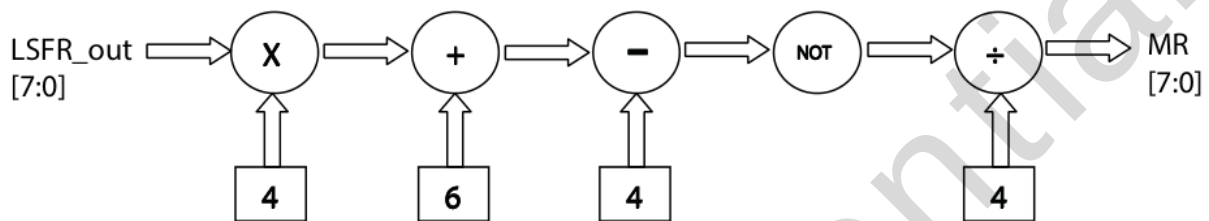


Figure 4-29 Diagram of ALU checker

Table 4-26 Monitoring module electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, $T_J = -40\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
MCU monitoring module						
t_{WD}	Timing window for watchdog (31)	10	-	60	ms	P_4.26.1
t_{VAR}	Variation of the watchdog window timing	60	75	90	ms	P_4.26.2

Notes:

31. The minimum timing window for the watchdog can be decreased down to a SPI timing access. Here, the range given in the [Table 4-27](#) shows the typical use case on an ABS application.

Sequence example to run the watchdog:

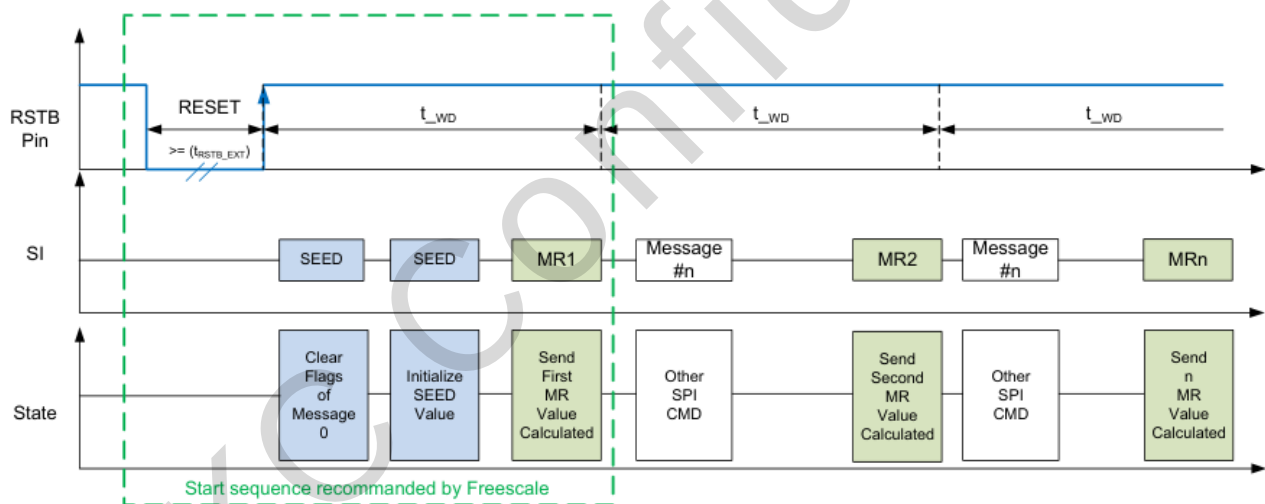
Check when the pin reset go high.

1st command: send the message 0: 0x00 to clear flags.

2nd command: send the message 0: 0x00 to reinitialize the seed value. Then, no need to send again the SEED value. 3rd command: message 18: send the corresponding MR value calculated due to the SEED value.

4th: send the message desired.

Note: The Message 18 (MR value) should be sent according to the t_{WD} timing

**Figure 4-30 Watchdog sequence example**

4.11 Analog to digital converter

ADC is referenced to VCC5 voltage and converts the voltage on 10 bits. It is used for reading the following voltage:

- Internal voltage supplies (VINT_A, VINT_D, Vcp-Vpwr).
- Average temperature of die, which is used by the temperature warning detection circuit (TEMP)

Table 4-27 ADC electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
MCU monitoring module						
Adc_err	Total Error	-6	—	+6	LSB	P_4.27.1
t _{CONV}	Conversion Time	—	6.6	7.5	us	P_4.27.2
t _{RFT}	Refresh Time	—	100	—	us	P_4.27.3
PDI						
APDI_LKG	Input Leakage Current	-2	—	+2	uA	P_4.27.4
APDI_CAP	Input Capacitance	—	—	30	pF	P_4.27.5
Internal voltage						
AD_VINT_D	V _{INT_D} Voltage (message 5 of the SPI table)	830	881	930	LSB	P_4.27.6
AD_VINT_A	V _{INT_A} Voltage (message 9 of the SPI table)	830	881	930	LSB	P_4.27.7
AD_VCP	V _{CP_VPWR} (message 8 of the SPI table) ⁽³⁴⁾	330	—	810	LSB	P_4.27.10
Temperature reading						
AD_TEMP25	Voltage at 25 °C (message 1 of the SPI table)	—	717	—	LSB	P_4.27.11
AD_DEV_TEMP	Deviation with 1.0 °C increment	—	-2	—	LSB/°C	P_4.27.12

Notes

32. NA

33. NA

34. ADC ratio = (VCP-VPWR)/4. 9.0 V < VPWR < 16 V

The ADC monitoring should be used to increase the safety level by doing a comparison with undervoltage, overvoltage, and overtemperature flag.

4.12 Supervision

Table 4-28 Reaction to supply fault and reset condition

Event	RSTB	WSx_HS, VSO, K-Line	LSDx	HD	PD	WLD	SPI
Normal mode: After RSTB rising edge, No fault	High	Normal	Normal	Normal	Normal	Normal	Normal
VINT_x undervoltage ⁽³⁷⁾	Low (output)	OFF	OFF	OFF	OFF	OFF	SPI register go to initial state Low except for V _{INT_UV} which is reset to 1. After first read of V _{INT_UV} , it is set back to 0.

Clock fail reset ⁽³⁷⁾	Low (output)	OFF	OFF	OFF	OFF	OFF	SPI registers go to initial state Low except for V_{INT_UV} unchanged & RST_clk which is set to 1. After first read of RST_clk, it is set back to 0.
DOSV undervoltage ⁽³⁷⁾	Low (output)	OFF	OFF	OFF after $t_{LSDx_HD_G}$	OFF	OFF	SPI register go to initial state except reset flag (V_{INT_UV} , VCC5_uv, DOSV_uv, RST_wd, RST_alu, RST_ext, RST_CLK) and hd_on ($t_{LSDx_HD_G}$ timing).

Table 4-29 Reaction to supply fault and reset condition (continued)

Event	RSTB	WSx_HS, VSO, K-Line	LSDx	HD	PD	WLD	SPI
VCC5 undervoltage ⁽³⁵⁾	Low (output)	OFF	OFF	OFF after $t_{LSDx_HD_G}$	OFF	OFF	SPI register go to initial state except reset flag (V_{INT_UV} , VCC5_uv, DOSV_uv, RST_wd, RST_alu, RST_ext, RST_CLK) and hd_on ($t_{LSDx_HD_G}$ timing).
Watchdog fault or ALU fault ⁽³⁵⁾	Low during RSTB_REC (output)	No effect	OFF	OFF after $t_{LSDx_HD_G}$	OFF	OFF	SPI register go to initial state except: reset flag (V_{INT_UV} , VCC5_uv, DOSV_uv, RST_wd, RST_alu, RST_ext, RST_CLK) WScfgx<1:0>, ocf_ws, > WSx_dis, ws_cnt_rst, ws_cnt_en - VSO_S<1:0> clear flag bits hd_on ($t_{LSDx_HD_G}$ timing). mr seed
External Reset	Low (input)	No effect	OFF	OFF after $t_{LSDx_HD_G}$	OFF	OFF	Same SPI behavior as Watchdog fault or ALU fault except seed. SPI register go to initial state except: reset flag (V_{INT_UV} , VCC5_uv, DOSV_uv, RST_wd, RST_alu, RST_ext, RST_CLK) WScfgx<1:0>, ocf_ws, > WSx_dis, ws_cnt_rst, ws_cnt_en - VSO_S<1:0> clear flag bits hd_on ($t_{LSDx_HD_G}$ timing). mr Watchdog circuit Seed register (written by SPI write Message #0) is reset by External reset event and impossible to write to unless External reset flag is cleared. In the case of External reset, two identical SPI write Message #0 should be executed to reinitialize Seed register. First SPI Message #0 clears External reset flag and second SPI Message #0 finally reinitializes the Seed register. In application, writing seed is preferred after checking if RST_ext is 1. (Only External Reset flag is to block writing Seed). Refer to the command Watchdog sequence, Figure 4-30
VPWR overvoltage	No effect	Fvpwr_ac t	OFF	OFF after $t_{LSDx_HD_G}$	On	No effect	Following SPI registers go to initial state Low: LSDx duty cycle or current set point. hd_on pd_on (SPI bit reset but output is forced ON by load dump logic) Also following blocks are forced to OFF state only during the fault duration (not latched): WSx_HS, VSO, K-line.

VPWR undervoltage	No effect	Depends on SPI bit Fvpwr_act	OFF	OFF after tLSDx_HD_G	OFF	No effect	Following SPI registers go to initial state Low: LSDx duty cycle or current set point. hd_on pd_on (SPI bit reset but output is forced ON by load dump logic) Also following blocks are forced to OFF state only during the fault duration (not latched): WSx_HS, VSO, K-line.
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Notes

35. State defines for the duration of the fault and the following reset recovery time period.

Restart conditions: SPI write message #0 has first to be executed to clear any reset or fault flags. Then new SPI command can be sent.

Table 4-30 Start point of reset recovery time

Fault Mode	Start Point of t_{RST_REC}
VINT_A or VINT_D_uv or VCC_uv or DOSV_uv	Come back normal voltage of all voltages
Watchdog or ALU fault	Fall edge of RSTB pin

4.12.1 Additional safety functions

4.12.1.1 VINT_A or VINT_D undervoltage supervision

The HE0400 uses an internal supply for analog functions (V_{INT_A}) and digital functions (V_{INT_D}). The supply voltage V_{INT_A} and V_{INT_D} are supervised for undervoltage. When the voltage becomes lower than each threshold $V_{INT_A_UV}$ and $V_{INT_D_UV}$, the RSTB pin is asserted low after detection filter time (t_{VINT}). This reset state continues until the voltage at the VINT raises again. And if VINT becomes higher than each threshold $V_{INT_A_UV}$ and $V_{INT_D_UV}$ for same filter time (t_{VINT}), the RSTB Pin goes high after reset recovery time (t_{RST_REC}) and the related flag of the SPI register is set to high. For stabilization the internal supply V_{INT_A} & V_{INT_D} requires external capacitors. Two band-gaps are included in the HE0400, one is for the voltage reference and the other is for the diagnostic. The V_{INT_A} and V_{INT_D} voltages are sending through SPI.

4.12.1.2 VCC5 supervision

See [Table 4-28](#) Reset condition and reaction

4.12.1.3 DOSV supervision

The supply voltage DOSV is supervised for undervoltage. When the voltage at pin DOSV becomes lower than $DOSV_{uv}$, the RST pin is asserted low after detection filter time (t_{VDUV}). This reset state continues until the voltage at pin DOSV raises again. And if DOSV becomes higher than ($DOSV_{uv}$) for same filter time (t_{VDUV}), the RSTB Pin goes high after reset recovery time (t_{RST_REC}) and the related flag of the SPI register is set to high. The pin P53_CFG decides the DOSV pin undervoltage threshold

Pin	Condition	Description
P53_CFG	Short to GND	5.0 V DOSV undervoltage threshold
	Short to VCC5	3.3V DOSV undervoltage threshold

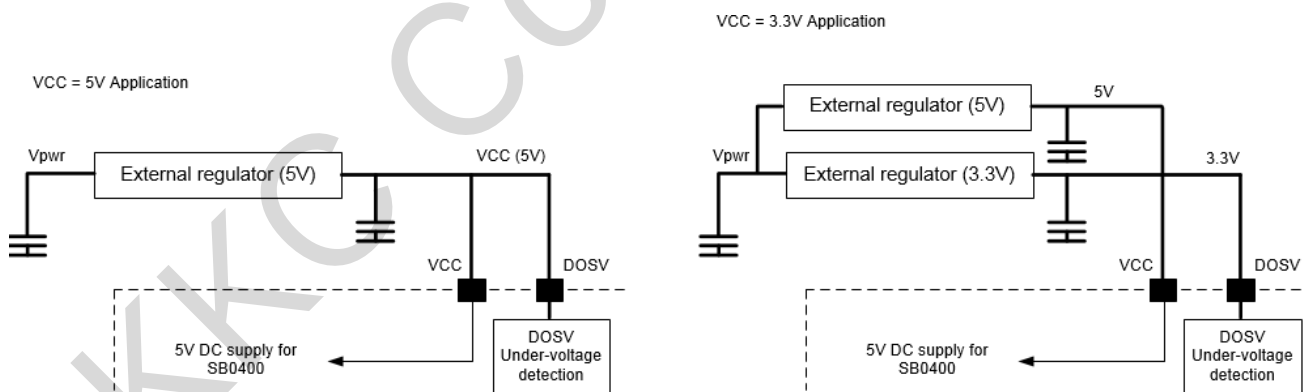


Figure 4-31 Configuration of VCC5 and DOSV for 5.0 V or 3.3 V application

4.12.1.4 VPWR supervision

4.12.1.5 Load dump protection

If a load dump occurs on the HD_D pin for a time period longer than t_{LD} , or V_{PWR_OV} for a time period longer than the overvoltage detection filter time (t_{VPWR_OV}), PD_G is activated to motor turn-on. PD_G is kept ON during t_{LD_ACT} after VPWR goes below the V_{PWR_OV} threshold & HD_D goes below HD_ov. If during t_{LD_ACT} , VPWR goes higher than the V_{PWR_OV} threshold, or HD_D is going higher than HD_ov then t_{LD_ACT} time is reset at its start value. When VPWR goes to its undervoltage condition during t_{LD_ACT} , PD_G turns off and the " t_{LD_ACT} " counter is set to the end value. The load dump function is disabled by the SPI command (LDact_dis), but the flag is displayed by LD bit thru the SPI.

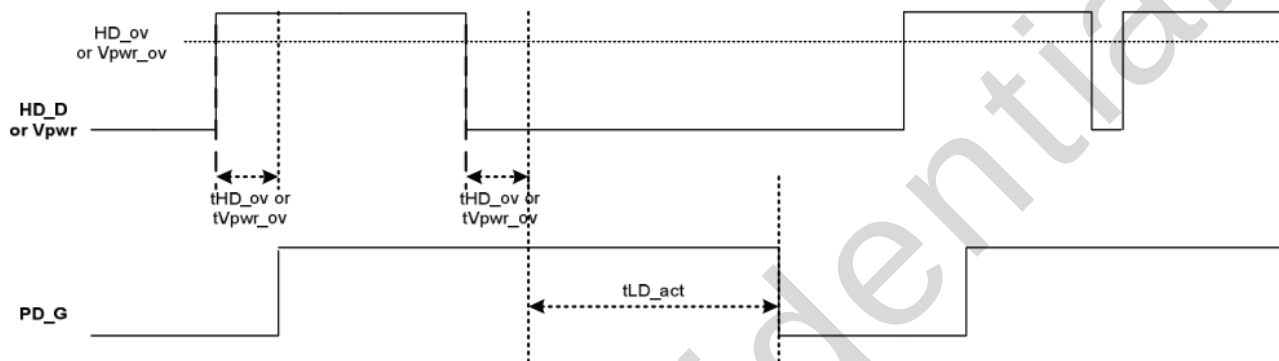


Figure 4-32 Timing of load dump detection

4.12.1.6 Charge pump

The charge pump generates a voltage of typically 12 V above the supply V_{PWR} . The charge pump voltage is intended for internal use only. No additional load must be connected to the CP pin. The charge pump requires a capacitor for energy storage and to cover transients. The voltage difference between CP and VPWR can be read by SPI.

4.12.1.7 Internal clock supervision (mismatch HE0400 MAIN-AUX CLK)

The HE0400 has two independent clock modules, one is the main supply clock to all HE0400 systems. The other monitors the main clock fault and if a fault is detected, the HE0400 resets with the RST_CLK function ([Table 4-28](#)). This function starts when RSTB is in a high state.

Mutual Supervision of Both Main and Auxiliary Clock

Clock monitoring continues to perform comparisons between the two clocks sources, CLK1 and CLK2. When everything is working correctly, both clocks are present and both have the same frequency of 14 MHz. If one of the clocks stops or if clocks are misaligned in frequency more than $\pm 25\%$ of 14 MHz ([Table 4-28](#)), an RSTB reset is generated ([Table 4-28](#)) and a SPI flag is reported (RST_CLK). The reset flag RST_CLK (same as other reset flags) is cleared in "clear on read" fashion, or in other words, the flag is cleared by a SPI Read command that reads the flag. In the case of a clock monitoring fault, the clock monitoring process restarts only after the clock monitoring flag (RST_CLK) is cleared on first SPI message.

If either CLK1 or CLK2 disappears indefinitely, the clock monitoring fault shows anywhere from T_1 to $2 \cdot T_2$. If clock frequencies are misaligned in more than $\pm 25\%$ of 14 MHz, the clock monitoring fault shows after a time delay of T_2 , as measured by the reference clock CLK1. The misaligned frequency detection error is measured in time window of T_2 and the measurement is based on CLK1 clock as reference, therefore if the CLK1 frequency changes, the time window T_2 cannot be guaranteed.

The HE0400 internal clock monitoring function can be disabled by the SPI command (StopCLK2), with no effect of functionality except the clock monitoring function, because CLK1 is activated, but CLK2 is deactivated. Frequency modulation can be controlled by the FM_amp and FM_EM bits (See [SPI and data register](#)). The SPI

command (FM_EN) enables the frequency modulated oscillator by two deviation frequency to spread the oscillator's energy over a wide frequency band. There are two kinds of deviation frequencies (350 kHz and 700 kHz), which are decided by the SPI command (FM_amp). This spreading decreases the peak electromagnetic radiation level and improves electromagnetic compatibility (EMC) performance.

If the "StopCLK2" is enabled and then reactivate (disabled), a reset clock (RST_clk) can be generated if the settling time is ≤ 1.0 ms (t_{CLK}). In this case, reset is detected during reset recovery time and the CLK_RST (reading message #0) flag should read in a normal condition.

4.12.1.8 Die temperature warning

The HE0400 has 1 temperature warning sensor in the cool place of the die. The threshold of temperature warning is 20 °C below overtemperature. In case of a temperature warning, outputs are not shutdown and the SPI-Bit shows the actual status at accessing time.

4.12.1.9 Ground supervision

GND-loss monitors the voltage between PGND (global reference GND) and GND_D. In case of a disconnection of GND_D vs. all other grounds (pin 3, 8, 29, 34, 36, and back side ground are soldered to ground), a detection GND_D disconnect as soon as the GND_D is higher than the threshold (V_{GL}) vs. others grounds, is reported through the flag FGND via the SPI register and set high after a filter time (t_{GL}).

1. Connection degraded (resistive path)
 - A GND_D vs other grounds $> V_{GL}$ but by having $V_{int_D} - GND_D > \text{min voltage required}$
 - B) SPI communication still possible, and the flag FGND will be at 1
2. Disconnection (open physically) during a sequence (in Normal mode), the logic embedded will be frozen, because the voltage $V_{int_D} - GND_D < \text{min voltage required}$
 - A No SPI communication is possible
 - B) If GND_D is reconnected normally, SPI communication recovers and the flag FGND will be at 1

Table 4-31 Electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Reset output HE0400 to MCU						
tRSTB_REC	Reset Recovery Time	-20%	45	20%	ms	P_4.31.1
Reset input MCU to HE0400						
tRSTB_EXT	External Reset Detection Filter Time - Filter on falling of RSTB pin. Mask shorter glitch.	1.0	—	4.0	us	P_4.31.2
tRST_MIN	Minimum External Reset Time (only for application) - System requirement (SW uC requirement; not tested)	8.0	10	—	us	P_4.31.3
DOSV undervoltage						
DOSVUV_5V DOSVUV_3P3	Undervoltage Reset Threshold at Shutdown (falling edge of DOSV) P53_cfg = Low (< 0.8 V) P53_cfg = High (> 2.0 V)	4.32 2.8	4.46 2.9	4.6 3.0	V	P_4.31.4
tDVUV	Undervoltage Reset Filter Time	—	T1	—	us	P_4.31.5
VCC5 undervoltage						
VCC5_uv	Undervoltage Threshold	4.32	4.46	4.6	V	P_4.31.6
tVCUV	Undervoltage Filter Time	—	T1	—	us	P_4.31.7
VCC5 supply						
I_VCC5 I_DOSV	Consumption Current VCC5 = 5.0 V; HD, PD = on; RSTB = high During SPI communication	— —	10 —	20 10	mA	P_4.31.8 P_4.31.9
Internal logic supply						
VINT_A	Internal Analog Voltage - ILOAD = -10 mA	4.1	4.3	4.5	V	P_4.31.10
VINT_D	Internal Digital Voltage - ILOAD = -10 mA	4.1	4.3	4.5	V	P_4.31.11
I_VINT_A I_VINT_D	Consumption current - guaranteed by design	— —	4.0 3.0	15 10	mA	P_4.31.13

Table 4-32 Electrical characteristics (continued)

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Internal logic supply undervoltage						
VINT_A_UV VINT_D_UV	Undervoltage Reset Threshold	3.6 3.6	3.72 3.72	3.8 3.8	V	P_4.32.1 P_4.32.2
tVINT	Undervoltage Reset Filter Time	—	T1	—	us	P_4.32.3
VPWR supply						
I_VPWR	Consumption Current - VPWR = 14 V, HD, PD = on, RSTB = high	—	10	20	mA	P_4.32.4
I_STBY_VPWR	Consumption Current at Sleep Mode - VCC5 = DOSV = 0 V, HD_D = PD_D = VPWR = 14 V	—	2.0	20	uA	P_4.32.5
I_STBY_VPWR	Global Leakage Current at Sleep Mode - I_STBY_VPWR + ILEAK_VSO + ILEAK_HD_DRN + ILEAK_PD_D + ILEAK_WLD condition: VPWR = HD_D = PD_D = WLD = 14 V, VCC5 = DOSV = 0 V	—	3.0	30	uA	P_4.32.6
VPWR & HD overvoltage, load dump activation						
VPWR_OV	VPWR Overvoltage Threshold (rising edge)	30	32	34	V	P_4.32.7
VPWR_OV_HYS	Overvoltage Detection Hysteresis - VPWR_OV(ON) = VPWR_OV(SHUTDOWN) - VPWR_OV_HYS	0.3	0.6	1.0	V	P_4.32.8
tVPWR_OV	Overvoltage Detection Filter Time - Both directions	—	T2	—	us	P_4.32.9
HD_OV	HD Overvoltage Threshold - Measured on HD_D	27.5	29.5	31.5	V	P_4.32.10
tHD_OV	HD Overvoltage Filter Time - Both direction	—	T2	—	us	P_4.32.11
tLD_ACT	Load Dump Activation Time - Apply after VPWR_OV & HD_OV conditions are removed	260	300	350	ms	P_4.32.12
VPWR undervoltage						
VPWR_UV	Undervoltage Shutdown Threshold (falling edge)	4.75	5.1	5.3	V	P_4.32.13
VPWR_UV_HYS	Undervoltage Detection Hysteresis - VPWR_UV(ON) = VPWR_UV(SHUTDOWN) - VPWR_UV_HYS	30	100	200	mV	P_4.32.14
tVPUV	Undervoltage Detection Filter Time	—	T2	—	us	P_4.32.15
Ground-loss detection						
V_GL	GND_d-loss detection threshold - Reference GND_Px	0.3	0.45	0.6	V	P_4.32.16
tGL	GND_d-loss detection filter time - Reference GND_Px	—	T2	—	us	P_4.31.17
Oscillator						
f_OSC	Main Oscillator Frequency	-7.0%	15	7.0%	MHz	P_4.31.18
eCLK	Mismatch HE0400 MAIN-AUX OSC CLK - enable VINT_X is normal voltage digital comparison between the 2 clocks.	-35	±25	35	%	P_4.31.19
tCLK	Mismatch OSC Filter Time (36)	T1	T2	2*T2	us	P_4.31.20
	Frequency Modulation Band 1 - FM_amp = 0	-30%	375	30%	kHz	P_4.31.21
	Frequency Modulation Band 2 - FM_amp = 1	-30%	750	30%	kHz	P_4.31.22

	Frequency Modulation Speed	-30%	110	30%	kHz	P_4.31.23
Overtemperature/temperature warning						
T _W	Temperature Warning Detection Threshold	150	165	180	°C	P_4.31.24
t _W	Temperature Warning Detection Filter Time	—	T2	—	us	P_4.31.25

Table 4-33 Electrical characteristics (continued)

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
Charge pump						
C _{CP}	External Charge Pump Capacitor (between VPWR pin and CP pin)- Tolerance < ±20% 20 V capacitor	—	220	—	nF	P_4.33.1
I _{INT_CP}	Charge Pump Internal Current Consumption - hd_on = pd_on = 1, LSD 100% duty cycle. VPWR = 14 V	10	350	400	uA	P_4.33.2
V _{CP}	Charge Pump Voltage - hd_on = pd_on = 1, LSD 100% duty cycle. (Minimum Charge pump Voltage is guaranteed by HD_G & PD_G spec. voltage)	See HD_G & PD_G spec.	VPWR+13	VPWR+15	V	P_4.33.3

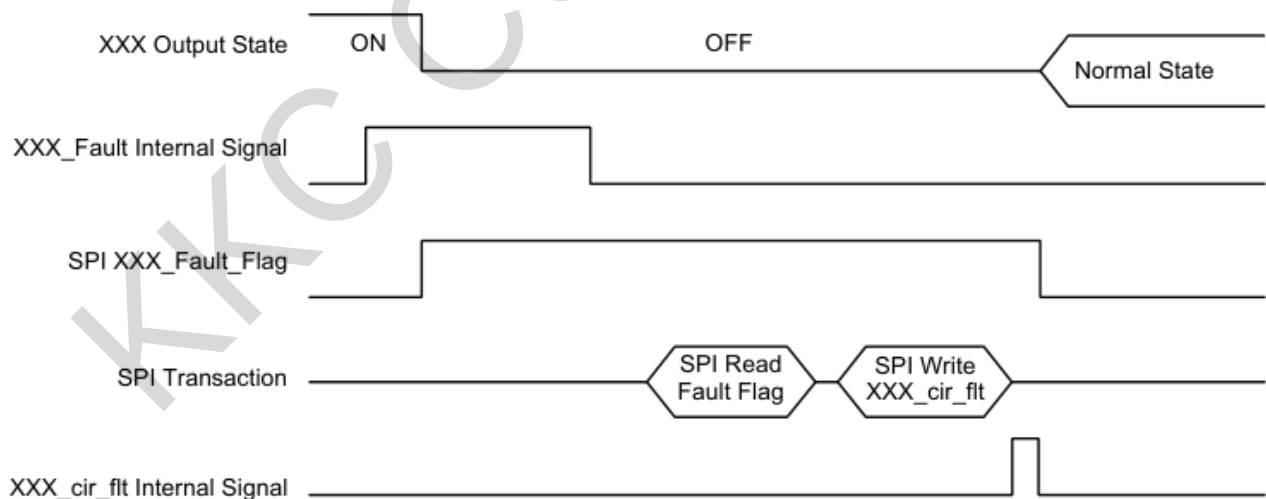
Timing

T1	Logic Time Base T1	14.4	18.2	22	us	P_4.33.4
T2	Logic Time Base T2	232	293	360	us	P_4.33.5

Notes

36. Parameter, *tolck* is decided by frequency checker and comparing two clocks. If either main clock or AUX clock frequency disappears longer than T1, HE0400 goes to reset by clock frequency checker and CLK_RST flag is detected. Meantime, comparing main clock and AUX clock is done during T2 and HE0400 is possible to go to reset every T2. Because measurement and reset activation are asynchronous, tCLK can reach 2*T2 in the worst case by comparing two clocks.

Write 1 to any xxx_clr_flg register creates a reset of the fault flag during 1 clock period after the SPI message. xxx_clr_flg automatically goes to "0" after 1 clock from fault flag reset.

**Figure 4-33 Timing diagram of xxx_clr_flg**

4.13 SPI and data register

4.13.1 Function description

The SPI serial interface has the following features:

- Full duplex, 4-wire synchronous communication
- Slave mode operation only
- Fixed SCLK polarity and phase requirements
- Fixed 16-bit command word
- SCLK operation up to 10.0 MHz

The Serial Peripheral Interface (SPI) is used to transmit and receive data synchronously with the MCU. Communication occurs over a full-duplex, four wire SPI bus. The HE0400 operates only as a slave device to the master, and requires four external pins; SI, SO, SCLK, and CSB. All words are 16 bits long and sent MSB first. The SPI simultaneously turns on the serial output SO and returns the MISO return bits. When receiving, valid data latches on the rising edge of each SCLK pulse. The serial output data is available on the rising edge of SCLK, and transitions on the falling edge of SCLK. The number of clock cycles occurring on the pin SCLK, while the CSB pin is asserted low, should be 16. If the number of clock pulses is not 16 or a parity fault, then the SPI MOSI data is ignored. The HE0400 takes even parity. On the next data read, the SO message "Fmsg" bit is set to 1. The Parity bit should be set such that the total number of 1-bits in the SPI word is even ($P = 0$ for an even number of 1-bits and $P = 1$ for an odd number). On the first SPI communication after reset, the read SO message is set to 1010101010101010.

The fault registers are double buffered. The first buffer layer latches a fault at the time the fault is detected. This inner layer buffer clears when the fault condition is no longer present and the fault bit communicates to the MCU with a MISO response. The second layer buffer latches the output of the inner layer buffer whenever the CSB pin transitions from low to high. The output of second layer buffer transfers to the shift register after the corresponding MOSI command is received from the MCU. The read data is the previous read data corresponding of the previous command sent.

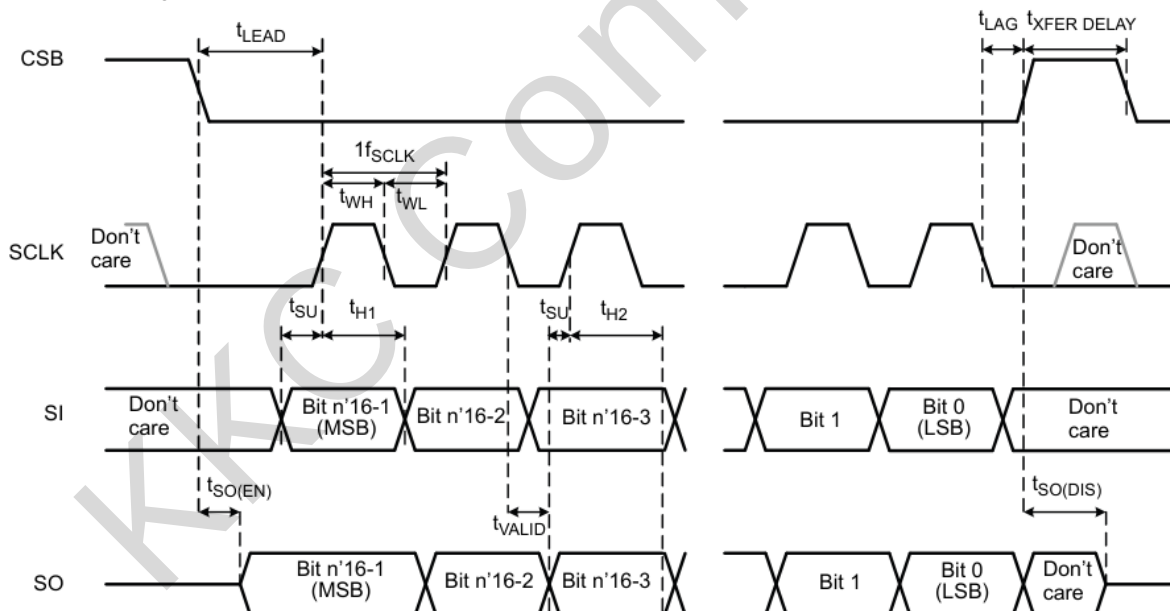


Figure 4-34 SPI timing diagram

Table 4-34 SPI and data register electrical characteristics

VPWR = 6.0 V to 20 V, VCC5 = 4.75 V to 5.25 V, DOSV = 3.13 V to 5.25 V, T_J = -40 °C to +150 °C, unless otherwise specified.

Symbol	Parameter	Min.	Typ.	Max.	Unit	Num
SPI interface timing ⁽³⁷⁾						
f _{SPI}	Recommended Frequency of SPI Operation - t _{SPI} = 1/f _{SPI}	—	—	10	MHz	P_4.34.1
t _{LEAD}	Falling Edge of CSB to Rising Edge of SCLK (Required Setup Time)	30	—	50	ns	P_4.34.2
t _{LAG}	Falling Edge of SCLK to Rising Edge of CSB (Required Setup Time)	30	—	50	ns	P_4.34.3
t _{XFER_DELAY}	No Data Time Between SPI Commands	300	—	—	ns	P_4.34.4
t _{WH}	High Time of SCLK	45	t _{SPI} /2	—	—	P_4.34.5
t _{WL}	Low Time of SCLK	45	t _{SPI} /2	—	—	P_4.34.6
t _{SU1}	SI to Rising Edge of SCLK (Required Setup Time)	15	—	—	ns	P_4.34.7
t _{SO(EN)}	Time from Falling Edge of CSB to SO Low-impedance	—	—	30	ns	P_4.34.8
t _{SO(DIS)}	Time from Rising Edge of CSB to SO High-impedance	—	—	30	ns	P_4.34.9
t _{VALID}	Time from Falling Edge of SCLK to SO Data Valid - 0.2xDOSV ≤ SO ≤ 0.8xDOSV, CL = 50 pF	0.0	—	30	ns	P_4.34.10

Notes

37. The inputs of the SPI module (SCLK, CSB, SI) are driven between 0 V and the DOSV voltage.

4.13.2 SPI message structure

addr #		Write										Read														
DEC	BI N	9	8	7	6	5	4	3	2	1	0	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0	00 00 0	SED<7:0>									HPD _sr	PD_m t_cf g	0011				dos v_uv	Vcc uv	Vin t_uv	RS T_cl k	RS T_ext	RS T_alu	RST wd	0	HPD _sr	PD_m t_cf g
1	00 00 1	X	X	X	X	X	X	X	l s d _s i n k _d i s	LDa ct _d is	0	Manufacturing data				X	X	X	X	X	X	X	l s d _s i n k _d i s	LDa ct _d is	X	
2	00 01 0	wscfg1< 1:0>		wscfg2< 1:0>		X	X	X	X	0	0	ws1 _o_c	ws1 _o_p	ws1 _o_t	ws2 _o_c	ws2 _o_p	ws2 _o_t	X	X	X	X	X	X	0	0	
3	00 01 1	ISO _fl t	ISO _de t	0	0	FM a m p	FM E N	Stop C L K2	WLD _en	PDI _en	OC F _pd	ASIC_CLK_CNT<7:0>									Vp wr ov	LD	Vpwr uv	FGN D	OT W	0
4	00 10 0	Fvp wr_ ac t	X	X	X	X	DF _ws	OCF _ws	VSO_S< 1:0>	vso _s el	Fvp wr_ ac t	VSO_S< 1:0>	vso _s el	TEMP<9:0>												
5	00 10 1	0	0	WS 2 dis	WS 1 dis	pd on	hd on	0	0	0	wld _on	PD _oc	HD _oc	0	0	VINT_D<9:0>										
6	00 11	X	X	VS O clr	ISO K clr	HD cl r	W S _cl r	0	WLD _cl	PD _cl	LSD _cl	wld _oc	wld _op	wld _ot	vds _wld	NOT USED										

	0			lt	lt				r_fit		lt												
7	00111	0	0	0	0	0	0	0	0	0	0	X	X	X	X	NOT USED							
8	01000	0	0	0	0	0	0	0	0	0	0	X	X	X	X	vcp_vpwr<9:0>							
9	01001	0	0	0	0	X	X	X	LF_PWM_14			Iso-pos 1	Iso-pos 2	X	X	VINT_A<9:0>							
14	01110	LSD1 duty cycle								0	0	lsd1_oc	lsd1_op	lsd1_ot	vds_LSD1	LSD1 duty cycle		ISO_K_oc	ISO_K_ot				
15	01111	LSD2 duty cycle								0	0	lsd2_oc	lsd2_op	lsd2_ot	vds_LSD2	LSD2 duty cycle		Iso-neg 1	Iso-neg 2				
16	10000	LSD3 duty cycle								0	0	lsd3_oc	lsd3_op	lsd3_ot	vds_LSD3	LSD3 duty cycle		X	X				
17	10001	LSD4 duty cycle								0	0	lsd4_oc	lsd4_op	lsd4_ot	vds_LSD4	LSD4 duty cycle		0	0				
18	10010	0	0	MR<7:0>							X	X	HD_lkg	ERR_CNT		AR<7:0>							
19	10011	0	0	0	ws_cnt_rst	ws_cnt_en	0	WSAI_S<1:0>		WS_S<1:0>		WS_CNT <7:0>				WS_CNT ov	0	WSAI_S<1:0>		WS_S<1:0>			
20	10100	0	0	0	0	0	0	0	0	0	0	W_S1_lkg	(WS11) data			Stop	Fail	Not legal	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0		(WS11) Manchester decoding result						code error		Number of not legal bits		

[illegible]

Notes

38. MSB(B15) of both write and read messages is parity bit, whereas only B14 of read message is Fmsg, which show previous write messagefault.
39. "X" means don't care bit

4.13.3 SPI message description

4.13.3.1 Message #0

Table 4-35 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					SED<7:0>								HPD_sr	PD_mt_cfg

Field	Bits	Description
P	15	The Parity bit should be set such that the total number of 1-bits in the SPI word is even (P = 0 for an even number of 1-bits and P = 1 for an odd number)
MSG_ID	14: 10	Message Identifier: 00000
SED<7:0>	09: 02	SEED value of Linear Feedback Shift Register
HPD_sr	01	Bit = 0 HSD&PD slew rate selection is slow (default mode)
		Bit = 1 HSD&PD slew rate selection is fast
PD_mt_cfg	00	Bit = 0 overcurrent masking time is mtPD_oc1 (default mode)
		Bit = 1 There is no overcurrent masking time

Table 4-36 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	0011				dosv_uv	Vcc 5_uv	Vint uv	RST_clk	RST_ext	RST_alu	RST_wd	0	HPD_sr	PD_mt_cfg

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
dosv_uv	09	Bit = 0 DOSV continues normal voltage
		Bit = 1 DOSV was less than DOSV undervoltage threshold longer than tDVUV
Vcc5 uv	08	Bit = 0 VCC5 continues normal voltage
		Bit = 1 VCC5 was less than VCC5_uv longer tVCUV
Vint uv	07	Bit = 0 Vint_D and Vint_A continues normal voltage
		Bit = 1 Vint_D or Vint_A voltage was low
RST_clk	06	Bit = 0 ASIC internal clock is okay
		Bit = 1 ASIC internal clock fault was detected.
RST_ext	05	Bit = 0 Normal
		Bit = 1 Reset from external (RSTB pin)
Field	Bits	Description
RST_alu	04	Bit = 0 Normal
		Bit = 1 Reset from monitoring module (ERR_CNT≥101)

RST_wd	03	Bit = 0	Normal
		Bit = 1	Valid MR wasn't updated within t_WD
HPD_sr	01	Bit = x	Feedback internal HPD_sr register value
PD_mt_cfg	00	Bit = x	Feedback internal PD_mt_cfg register value

4.13.3.2 Message #1

Table 4-37 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					X	X	X	X	X	X	X	lsd_sin k_dis	LDact_dis	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 00001
lsd_sink_dis	02	Bit = 0 LSD sink current for open detection is enabled (default mode)
		Bit = 1 LSD sink current for open detection is disabled
LDact_dis	03	Bit = 0 Load dump function is enabled (default mode)
		Bit = 1 The load dump function is disabled, but the flag is displayed by the LD bit through the SPI

Table 4-38 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	Manufacturing data				X	X	X	X	X	X	X	lsd_sin k_dis	LDact_dis	X

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
Manufacturing data	13:10	xxxx Reserved for FSL
ISD_SINK_DIS	02	Feedback of lsd_sink_dis
LDACT_DIS	01	Feedback of LDact_dis

4.13.3.3 Message #2

Table 4-39 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					wscfg1 <1:0>		wscfg2 <1:0>		X	X	X	X	0	0

Field	Bits	Description	
P	15	Parity bit	
MSG_ID	14:10	Message Identifier: 00010	
wscfg1<1:0>	09:08	Bit = 0X	Type 1 (The duty cycle of the standard sensor, default mode)
		Bit = 10	Type 2 (A pass-through mode for PWM-encoded sensor SPI transfers 5-bit WS1 data, stop, fail, and invalid information)
		Bit = 11	Type 3 (Pulse encoded sensors) SPI transfers 9-bit WS1 data, code error, legal bit[2:0]
wscfg2<1:0>	07:06	WS2 sensor type can be varied by wscfg2<1:0>	

Table 4-40 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	ws1_oc	ws1_op	ws1_ot	ws2_oc	ws2_op	ws2_ot	X	X	X	X	X	X	0	0

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
wsx_oc	13, 10	Bit = 0	Normal
		Bit = 1	Overcurrent shut down of WS1~2
wsx_op	12, 09	Bit = 0	Normal
		Bit = 1	Open Load/ short to battery detection of WS1~2
wsx_ot	11, 08	Bit = 0	Normal
		Bit = 1	Overtemperature shutdown of WS1~2

4.13.3.4 Message #3

Table 4-41 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					ISO_fit	ISO_det	0	0	FM_amp	FM_EN	StopCLK2	WLD_en	PDI_en	OCF_pd

Field	Bits	Description	
P	15	Parity bit	
MSG_ID	14:10	Message Identifier: 00011	
ISO_fit	09	Bit = 0	When ISO_det is 1 and ISO_pos or ISO_neg is detected, WSOx is latched during 50 usec

		Bit = 1	When ISO_det is 1 and ISO_pos or ISO_neg is detected, WSOx is latched during 100 usec
ISO_det	08	Bit = 0	—
		Bit = 1	wsox maintains the previous value during tiso_ft
FM_amp	05	Bit = 0	Frequency modulation band 1
		Bit = 1	Frequency modulation band 2

Field	Bits	Description	
FM_EN	04	Bit = 0	Frequency of Main/Aux oscillator clocks is fixed
		Bit = 1	Frequency of Main/Aux oscillator clock is modulated by the frequency defined by FM_amp
StopCLK2	03	Bit = 0	ASIC internal clock monitoring function is enabled
		Bit = 1	ASIC internal clock monitoring function is disabled
WLD_en	02	Enable the control of WLD	
PDI_en	01	Allow to control the motor pump pre-driver directly through the PDI pin.	
OCF_pd	00	Overcurrent filter time of pump driver is selectable by OCF pd	

Table 4-42 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	ASIC_CLK_CNT<7:0>								VPWR OV	LD	VPWR UV	FGND	OTW	0

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
ASIC_CLK_CNT<7:0>	13:06	Monitoring result from ASIC internal clock: Freewheeling counter is an 8-bit counter and increase every 36.5 μ s (512 cycle from 14 MHz clock) (10% accuracy with margin). This can be used for monitoring internal clock of ASIC by reading difference of ASIC_CLK_CNT<7:0> value between SPI accesses. 7.0 ms period for example should give ~192 LSB difference (192x36.5 μ s = 7.0 ms)	
Vpwr ov	05	Bit = 0	Normal
		Bit = 1	VPWR overvoltage
LD	04	Bit = 0	Normal
		Bit = 1	Load dump detection
Vpwr uv	03	Bit = 0	Normal
		Bit = 1	VPWR undervoltage
FGND	02	Bit = 0	Normal
		Bit = 1	GND_D loss detection
OTW ov	01	Bit = 0	Normal
		Bit = 1	Overtemperature warning

4.13.3.5 Message #4

Table 4-43 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					Fvpwr_act	X	X	X	X	DF_ws	OCF_ws	VSO_S<1:0>	vso_sel	

Field	Bits	Description	
P	15	Parity bit	
MSG_ID	14:10	Message Identifier: 00100	
Fvpwr_act	09	Bit = 0	WSx_HS, VSO, and K-line are OFF in case of VPWR undervoltage or overvoltage condition
		Bit = 1	WSx_HS, VSO, and K-line are normally operated in case of VPWR undervoltage or overvoltage condition

Field	Bits	Description	
DF_ws	04	Bit = 0	WSOx has no digital filter time
		Bit = 1	WSOx has digital filter time (tF_WS)
OCF_ws	03	Overcurrent filter time of WSS is controllable	
VSO_S<1:0>	02:01	Bit = 00	WSI1 input is selected
		Bit = 01	WSI2 input is selected
vso_sel	00	Bit = 0	VSO send vehicle speed output from WSOx which is chosen by VSO_S<1:0> (default mode)
		Bit = 1	VSO_IN can control High voltage output

Table 4-44 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	Fvpwr_act	VSO_S<1:0>	vso_sel	TEMP<9:0>										

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
Fvpwr_act	13	Feedback of Fvpwr_act	
VSO_S<1:0>	12:11	Feedback of VSO_S<1:0>	
vso_sel	10	Feedback of vso_sel	
TEMP<9:0>	09:00	Die temperature (10-bit)	

4.13.3.6 Message #5

Table 4-45 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	WS2_dis	WS1_dis	pd_on	hd_on	0	0	0	wld_on

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 00101
WS2_dis	07	Bit = 0 WS2 high-side switch is ON (default mode)
		Bit = 1 WS2 high-side switch is OFF
WS1_dis	06	Bit = 0 WS1 high-side switch is ON (default mode)
		Bit = 1 WS1 high-side switch is OFF
pd_on	05	Bit = 0 Pump Motor driver is off (default mode), Pump motor driver can turn on during load dump
		Bit = 1 Pump Motor driver is on
hd_on	04	Bit = 0 High-side driver is off (default mode)
		Bit = 1 High-side driver is on
wld_on	00	Bit = 0 WLD is off state
		Bit = 1 WLD turn on

Table 4-46 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	PD_oc	HD_oc	0	0	VINT_D<9:0>									

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
PD_oc	13	Bit = 0 Normal
		Bit = 1 Overcurrent shutdown of pump motor driver
HD_oc	12	Bit = 0 Normal
		Bit = 1 Overcurrent shutdown of high-side driver
VINT_D<9:0>	09:00	Internal digital voltage supply (10-bit)

4.13.3.7 Message #6

Table 4-47 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					X	X	VSO clr_fit	ISOK clr_fit	HD_clr flt	WS_clr flt	0	WLD_clr r_fit	PD_clr flt	LSD_clr flt

Field	Bits	Description	
P	15	Parity bit	
MSG_ID	14:10	Message Identifier: 00110	
VSO_clr_fit	07	Bit = 0	Overcurrent and overtemperature flags of VSO are conserved (default mode)
		Bit = 1	Clear overcurrent and overtemperature flags of VSO
ISOK_clr_fit	06	Bit = 0	ISOK_oc and ISOK_ot are conserved (default mode)
		Bit = 1	Clear ISOK_oc and ISOK_ot
HD_clr_fit	05	Bit = 0	HD_oc and HD_lkg are conserved (default mode)
		Bit = 1	Clear HD_oc, HD_lkg
WS_clr_fit	04	Bit = 0	All wsx_oc and wsx_ot are conserved (default mode)
		Bit = 1	Clear all wsx_oc and wsx_ot
WLD_clr_fit	02	Bit = 0	WLD_oc and WLD_ot are conserved (default mode)
		Bit = 1	Clear WLD_oc and WLD_ot
PD_clr_fit	01	Bit = 0	PD_oc is conserved (default mode)
		Bit = 1	Clear PD_oc
LSD_clr_fit	00	Bit = 0	All LSDx_oc and LSDx_ot are conserved (default mode)
		Bit = 1	All LSDx_oc and LSDx_ot are cleared

Table 4-48 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	wld_oc	wld_op	wld_ot	vds_wld	NOT USED									

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
wld_oc	13	Bit = 0	Normal
		Bit = 1	Overcurrent shutdown of warning lamp
wld_op	12	Bit = 0	Normal
		Bit = 1	Open load detection of warning lamp
wld_ot	11	Bit = 0	Normal
		Bit = 1	Overtemperature shutdown of warning lamp
vds_wld	10	Bit = 0	Normal
		Bit = 1	Vds detection of warning lamp (information only)

4.13.3.8 Message #7

Table 4-49 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	0	0	0	0	0	0	0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 00101

Table 4-50 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	X	X	X	X	NOT USED									

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer

4.13.3.9 Message #8

Table 4-51 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	0	0	0	0	0	0	0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14: 10	Message Identifier: 01000

Table 4-52 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	X	X	X	X	vcp_vpwr<9:0>									

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid.
		Bit = 1 Parity bit is not correct. Error detected during previous transfer.
vcp_vpwr<9:0>	09:00	Internal Charge Pump voltage supply (10-bit)

5.12.1.1 Message #9

Table 4-53 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	0	0	X	X	X	LF_PWM_14		

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 01001
LF_PWM_14	02:00	Bit = 000 Output PWM frequency of LSD(1~4)= 3.9kHz
		Bit = 001 Output PWM frequency of LSD(1~4)=4.5kHz
		Bit = 010 Output PWM frequency of LSD(1~4)=5.0kHz
		Bit = 011 Output PWM frequency of LSD(1~4)=4.2kHz
		Bit = 100 Output PWM frequency of LSD(1~4)=3.6kHz
		Bit = 101 Output PWM frequency of LSD(1~4)=3.4kHz
		Bit = 110 Output PWM frequency of LSD(1~4)=3.2kHz
		Bit = 111 Output PWM frequency of LSD(1~4)= 3.0kHz

Table 4-54 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	ISO_pos1	ISO_pos2	X	X	VINT_A<9:0>									

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
ISO_pos1	13	Bit = 0	no fault detected on WSS1_HS
		Bit = 1	It is detected that WSS1_HS current is lower than lws_open
ISO_pos2	13	Bit = 0	no fault detected on WSS2_HS
		Bit = 1	It is detected that WSS2_HS current is lower than lws_open
VINT_A<9:0>	09:00	Internal Analog voltage supply (10-bit)	

4.13.3.10 Message #14

Table 4-55 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					LSD1 duty cycle								0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 01110
LSD1 duty cycle	09:02	LSD1 PWM duty cycle

Table 4-56 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	lsd1_oc	lsd1_op	lsd1_ot	vds_LSD1	LSD1 duty cycle								ISOK_oc	ISOK_ot

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
lsd1_oc	13	Bit = 0 Normal
		Bit = 1 Overcurrent shut down of LSD1
lsd1_op	12	Bit = 0 Normal
		Bit = 1 Open load detection of LSD1
lsd1_ot	11	Bit = 0 Normal
		Bit = 1 Overtemperature shutdown of LSD1
vds_LSD1	10	Bit = 0 Normal
		Bit = 1 Vds detection of LSD1 (information only)
LSD1 duty cycle	09:02	Read programmed PWM duty cycle (to check SPI write) (8 bits for the range to 100%)
ISOK_oc	01	Bit = 0 Normal
		Bit = 1 Overcurrent shutdown of ISOK
ISOK_ot	00	Bit = 0 Normal
		Bit = 1 Overtemperature shut down of ISOK

4.13.3.11 Message #15

Table 4-57 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					LSD2 duty cycle								0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 01111
LSD2 duty cycle	09:02	LSD2 PWM duty cycle

Table 4-58 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	lsd2_oc	lsd2_op	lsd2_ot	vds_LSD2	LSD2 duty cycle								ISO_neg1	ISO_neg2

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
lsd2_oc	13	Bit = 0 Normal
		Bit = 1 Overcurrent shut down of LSD2
lsd2_op	12	Bit = 0 Normal
		Bit = 1 Open load detection of LSD2
lsd2_ot	11	Bit = 0 Normal
		Bit = 1 Overtemperature shutdown of LSD2
vds_LSD2	10	Bit = 0 Normal
		Bit = 1 Vds detection of LSD2 (information only)
LSD2 duty cycle	09:02	Read programmed PWM duty cycle (to check SPI write) (8 bits for the range to 100%)
ISO_neg1	01	Bit = 1 It detected WSS1_HS current is higher than WS_oc
ISO_neg2	00	Bit = 1 It detected WSS2_HS current is higher than WS_oc

4.13.3.12 Message #16

Table 4-59 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					LSD3 duty cycle								0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 10000
LSD3 duty cycle	09:02	LSD3 PWM duty cycle

Table 4-60 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	lsd3_oc	lsd3_op	lsd3_ot	vds_LSD3	LSD3 duty cycle								X	X

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
lsd3_oc	13	Bit = 0 Normal
		Bit = 1 Overcurrent shut down of LSD3
lsd3_op	12	Bit = 0 Normal
		Bit = 1 Open Load detection of LSD3
lsd3_ot	11	Bit = 0 Normal
		Bit = 1 Overtemperature shutdown of LSD3
vds_LSD3	10	Bit = 0 Normal
		Bit = 1 Vds detection of LSD3 (information only)
LSD3 duty cycle	09:02	Read programmed PWM duty cycle (to check SPI write) (8 bits for the range to 100%)

4.13.3.13 Message #17

Table 4-61 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					LSD4 duty cycle								0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 10001
LSD4 duty cycle	09:02	LSD4 PWM duty cycle

Table 4-62 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	lsd4_oc	lsd4_op	lsd4_ot	vds_LSD4	LSD4 duty cycle								0	0

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
lsd4_oc	13	Bit = 0 Normal
		Bit = 1 Overcurrent shut down of LSD4
lsd4_op	12	Bit = 0 Normal
		Bit = 1 Open load detection of LSD4
lsd4_ot	11	Bit = 0 Normal
		Bit = 1 Overtemperature shutdown of LSD4
vds_LSD4	10	Bit = 0 Normal
		Bit = 1 Vds detection of LSD4 (information only)
LSD4 duty cycle	09:02	Read programmed PWM duty cycle (to check SPI write) (8 bits for the range to 100%)

4.13.3.14 Message #18

Table 4-63 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	MR<7:0>							

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 10010
MR<7:0>	07:00	Monitoring result of MCU

Table 4-64 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	X	X	HD_lkg	ERR_CNT			AR<7:0>							

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
HD_lkg	11	Bit = 0 Normal
		Bit = 1 High-side driver leakage detected
ERR_CNT	10:08	3 bit Error counter value of monitoring logic
AR<7:0>	07:00	Monitoring result of ASIC

4.13.3.15 Message #19

Table 4-65 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	0	ws_cnt_rst	ws_cnt_en	0	WSAI_S<1:0>		WS_S<1:0>	

Field	Bits	Description		
P	15	Parity bit		
MSG_ID	14:10	Message Identifier: 10011		
ws_cnt_rst	06		Bit = 0	Bit = 1
ws_cnt_en	05	Bit = 0	Counter keeps previous value	Counter reset stop as 0 when ws_cnt_rst=1
		Bit = 1	Counter step up	Counter reset start to run
WSAI_S<1:0>	03:02	Bit = 00	Transfer Type 3 sensor additional information of WS1_HS to WSAI	
		Bit = 01	Transfer Type 3 sensor additional information of WS2_HS to WSAI	
WS_S<1:0>	01:00	Bit = 00	WSO_1 is selected for counter source	
		Bit = 01	WSO_2 is selected for counter source	

Table 4-66 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	WS_CNT <7:0>								WS_CNT ov	0	WSAI_S <1:0>		WS_S<1:0>	

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
WS_CNT <7:0>	13:06	8-bit wheel speed counter	
WS_CNT ov	05	Bit = 0	Wheel speed counter is less than 256
		Bit = 1	Wheel speed counter exceed 256 (information only)
WSAI_S<1:0>	03:02	Feedback of WSAI_S<1:0>	
WS_S<1:0>	03:02	Feedback of WS_S<1:0>	

4.13.3.16 Message #20

Table 4-67 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	0	0	0	0	0	0	0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 10100

Table 4-68 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	WS1_lkg	(WSI1) data					Stop	Fail	Invalid	0	0	0	0	0
			(WSI1) Manchester decoding result									code error	Number of invalid bits		

Type 1 & 2 Wheel speed sensor (wscfg1 #11)

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
WS1_lkg	13	Bit = 0 WS1 input has no leakage current fault
		Bit = 1 WS1 input has leakage current fault
(WSI1) data	12:08	WS1 additional information (high level pulse length)
Stop	07	Bit = 0 WS1 not in stop condition
		Bit = 1 WS1 stop in condition
Fail	06	Bit = 0 WS1 does not fail
		Bit = 1 WS1 fails
Invalid	05	Bit = 0 WS1 is valid ($t_L > t_{th8}$)
		Bit = 1 WS1 is invalid ($t_L < t_{th8}$)

Type 3 Wheel speed sensor (wscfg1 =11)

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
WS1_lkg	13	Bit = 0	WS1 input has no leakage current fault
		Bit = 1	WS1 input has leakage current fault
(WS1) Manchester decoding result	12:04	WS1 additional information (Manchester decoding result) Data bit "0" is located in MSB (bit address 12)	
code error	03	Bit = 0	Manchester decoding succeeds
		Bit = 1	Manchester decoding has error (No edge is detected in detection window)
Number of invalid bits	02:00	Bit = 000	Number of invalid bits are 0 (high speed mode)
		Bit = 001	Number of invalid bits are 1 (high speed mode)
		Bit = 010	Number of invalid bits are 2 (high speed mode)
		Bit = 011	Number of invalid bits are 3 (high speed mode)
		Bit = 100	Number of invalid bits are 4 (high speed mode)
		Bit = 101	Number of invalid bits are 5 (high speed mode)
		Bit = 110	Number of invalid bits are 6 (high speed mode)
		Bit = 111	Number of invalid bits are 7 (high speed mode)

4.13.3.17 Message #21

Table 4-69 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	0	0	0	0	0	0	0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 10101

Table 4-70 Read message (depends on wscfg2<1:0>)

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	WS2_lkg	(WSI2) data					Stop	Fail	Invalid	0	0	0	0	0
			(WSI2) Manchester decoding result									code error	Number of invalid bits		

Type 1 & 2 wheel speed sensor (wscfg2 #11)

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
WS2_lkg	13	Bit = 0 WS2 input has no leakage current fault
		Bit = 1 WS2 input has leakage current fault
(WSI2) data	12:08	WS2 additional information (high level pulse length)

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Type 1 & 2 wheel speed sensor (wscfg2 #11)

Field	Bits	Description	
Stop	07	Bit = 0	WS2 not in stop condition
		Bit = 1	WS2 stop in condition
Fail	06	Bit = 0	WS2 does not fail
		Bit = 1	WS2 fails
Invalid	05	Bit = 0	WS2 is valid ($t_L > t_{th8}$)
		Bit = 1	WS2 is invalid ($t_L < t_{th8}$)

Type 3 wheel speed sensor (wscfg2 =11)

Field	Bits	Description	
P	15	Parity bit	
Fmsg	14	Bit = 0	Parity bit is correct. Previous transfer was valid
		Bit = 1	Parity bit is not correct. Error detected during previous transfer
WS2_lkg	13	Bit = 0	WS2 input has no leakage current fault
		Bit = 1	WS2 input has leakage current fault
(WSI2) Manchester decoding result	12:04	WS2 additional information (Manchester decoding result) Data bit "0" is located in MSB (bit address 12)	
code error	03	Bit = 0	Manchester decoding succeeds
		Bit = 1	Manchester decoding has error (No edge is detected in detection window)
Number of invalid bits	02:00	Bit = 000	Number of invalid bits are 0 (high speed mode)
		Bit = 001	Number of invalid bits are 1 (high speed mode)
		Bit = 010	Number of invalid bits are 2 (high speed mode)
		Bit = 011	Number of invalid bits are 3 (high speed mode)
		Bit = 100	Number of invalid bits are 4 (high speed mode)
		Bit = 101	Number of invalid bits are 5 (high speed mode)
		Bit = 110	Number of invalid bits are 6 (high speed mode)
		Bit = 111	Number of invalid bits are 7 (high speed mode)

4.13.3.18 Message #26

Table 4-71 Write message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	MSG_ID					0	0	0	0	0	0	0	0	0	0

Field	Bits	Description
P	15	Parity bit
MSG_ID	14:10	Message Identifier: 11010

Table 4-72 Read message

B15	B14	B13	B12	B11	B10	B09	B08	B07	B06	B05	B04	B03	B02	B01	B00
P	Fmsg	vso_oc	vso_op	vso_ot	vds_vso	X	X	X	X	X	X	X	X	X	X

Field	Bits	Description
P	15	Parity bit
Fmsg	14	Bit = 0 Parity bit is correct. Previous transfer was valid
		Bit = 1 Parity bit is not correct. Error detected during previous transfer
vso_oc	13	Bit = 0 Normal
		Bit = 1 Overcurrent shutdown of VSO
vso_op	12	Bit = 0 Normal
		Bit = 1 Open load detection of VSO
vso_ot	11	Bit = 0 Normal
		Bit = 1 Overtemperature shutdown of VSO
vds_vso	10	Bit = 0 Normal
		Bit = 1 Overcurrent shutdown of VSO

4.13.3.19 5.13.3.20 Message #7, 8, 10, 11, 12, 13, 22, 23, 24, 25

Reserved for tests production.

Notes:

- 40. Message #7 & #8 can be only read.

5 Typical applications

5.1 Application diagrams

This typical application presents an ABS application schematic using the HE0400.

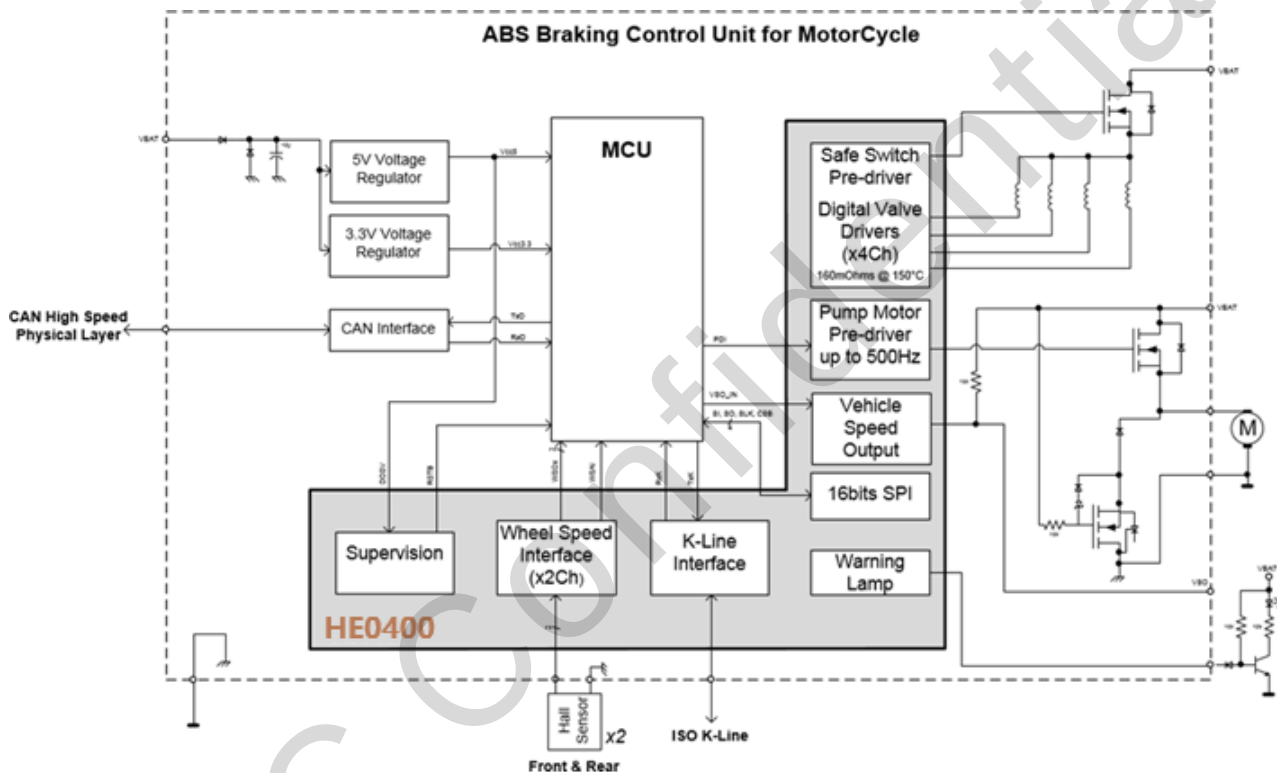
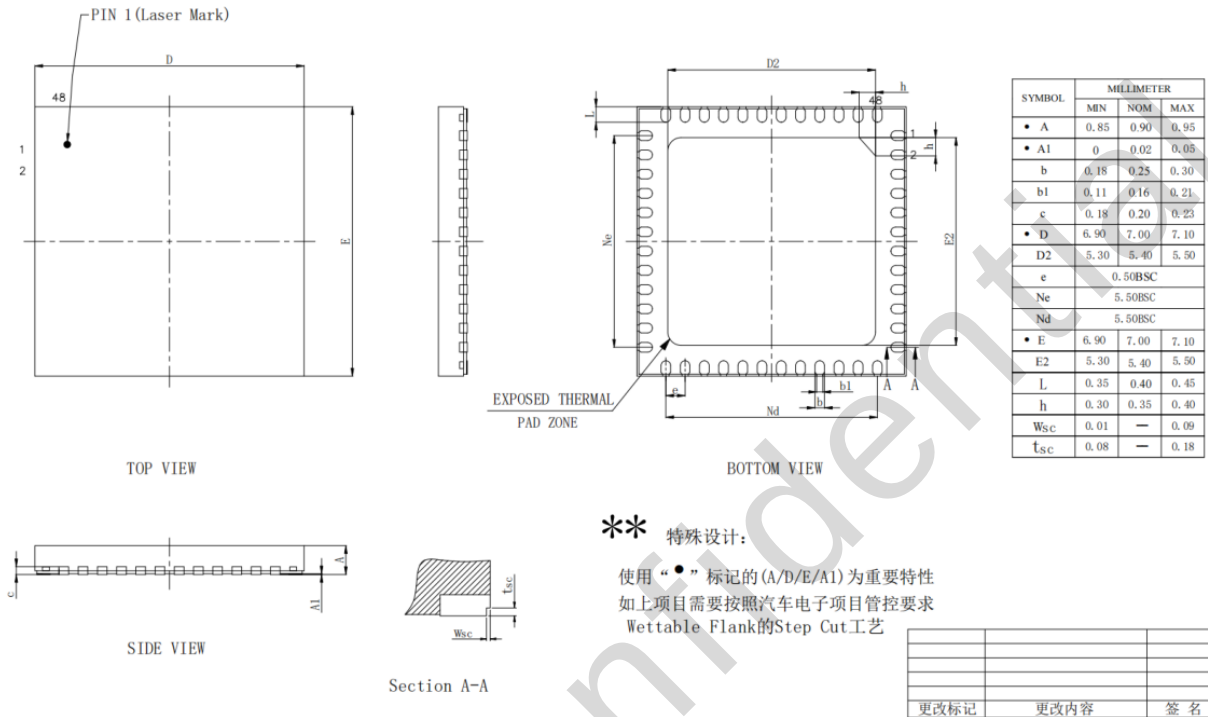


Figure 5-1 Motorcycle ABS control unit for two wheel simplified diagram

6 Packaging

6.1 Package mechanical dimensions

G



7 Revision history

Revision	Date	Description of changes
0.0.1	11/2025	初版
0.0.2	01/2025	调整格式若干
	01/2025	更新原手册参数单位错误
0.0.3	02/2025	更新系列参数&功能指标
0.0.4	03/2025	更新系列参数&功能指标